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CAPSTONE PROJECT PROPOSAL

LEXSHIELD: Integrated Cybersecurity and Legal Compliance Platform for Law Firms

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CHAPTER 1

INTRODUCTION

1.1 Background of the Study

Technology has changed the way law firms manage their everyday work. Many firms now use computers, cloud storage, email, and online communication to create, store, and share legal documents. Lawyers can also communicate with clients, access case files, and work remotely using digital tools. These technologies make work faster, more organized, and more efficient. However, as law firms become more dependent on technology, they also face greater risks from cyber threats such as phishing, ransomware, malware, hacking, and data breaches that can compromise sensitive information (Verizon, 2025). Those attacks can cause important legal documents to be lost and may expose private client information without permission.

Law firms handle a large amount of sensitive and confidential data, including personal information, financial records, contracts, court documents, and legal strategies. Clients trust law firms to keep this information safe and private. If confidential data is stolen, altered, or leaked, it may harm clients, damage ongoing legal cases, and affect the reputation of the law firm. It may also result in financial losses and legal liabilities. Because of these risks, protecting digital information has become one of the most important responsibilities of every law firm.

Cybersecurity is the practice of protecting computers, networks, software, and digital information from unauthorized access, cyberattacks, and data theft. According to the National Institute of Standards and Technology (NIST, 2024), cybersecurity helps organizations protect the confidentiality, integrity, and availability of information. Confidentiality ensures that only authorized individuals can access sensitive data. Integrity protects information from being

changed or damaged without permission, while availability ensures that information is accessible whenever it is needed. By implementing effective cybersecurity measures, organizations can reduce risks, protect valuable information, and continue their operations without major disruptions (National Institute of Standards and Technology [NIST], 2024).

Aside from protecting their computer systems, law firms must also comply with laws and regulations related to data privacy and information security. In the Philippines, the Data Privacy Act of 2012 (Republic Act No. 10173) requires organizations to protect personal information and apply reasonable organizational, physical, and technical security measures (Republic of the Philippines, 2012). The law also promotes transparency, accountability, and responsible handling of personal data. According to the National Privacy Commission (2022), organizations that fail to comply with these requirements may face administrative penalties, legal sanctions, financial losses, and damage to their professional reputation. Since law firms regularly collect and process personal information from clients, employees, and other parties, compliance with these regulations is an important part of their daily operations (National Privacy Commission, 2022).

Many law firms use different software applications to manage cybersecurity and legal compliance. For example, one system may be used for antivirus protection, another for firewall monitoring, another for document storage, and another for compliance tracking. While these tools can perform their individual functions effectively, using multiple systems often creates challenges. Employees may need to switch between several applications, making work more time-consuming and increasing the possibility of missing security alerts, software updates, or compliance deadlines. Managing different platforms may also require additional training, higher costs, and more effort from both legal and information technology personnel (International Organization for Standardization [ISO], 2022; Ries, 2023).

Because of these challenges, there is a growing need for an integrated platform that combines cybersecurity monitoring and legal compliance management into a single system. Having one centralized platform can help law firms monitor security threats, organize compliance activities, manage important documents, and generate reports more efficiently. It can also simplify daily operations, reduce human error, improve response to cybersecurity incidents, and support compliance with data privacy regulations (ISO, 2022).

Thus, this capstone project proposes the development of **LEXSHIELD: Integrated Cybersecurity and Legal Compliance Platform for Law Firms**. The proposed platform aims to provide law firms with a centralized solution that combines cybersecurity protection and legal compliance management in one system. It will help monitor security risks, organize compliance requirements, manage legal and security-related documents, and strengthen the protection of sensitive client information. By combining cybersecurity and legal compliance in one platform, the proposed system aims to help law firms work more efficiently and stay organized. It can help protect sensitive client information, make it easier to follow legal requirements, reduce cybersecurity risks, and improve the overall safety and reliability of the firm's daily operations. LEXSHIELD, law firms can better protect confidential information, respond to evolving cyber threats, and maintain the trust and confidence of their clients.

1.2 Statement of the Problem

The increasing use of digital technology in law firms has improved the way legal services are delivered and managed. Law firms now rely on computers, online storage, electronic documents, and communication platforms to handle important information and communicate with clients. However, the use of these technologies also creates new challenges, especially in protecting confidential information and maintaining compliance with data privacy laws.

Many Philippine law firms still face difficulties in managing cybersecurity and legal compliance because existing solutions are often separated into different systems. Some firms use security tools to protect their devices and networks, while compliance tasks are handled manually through documents, spreadsheets, or separate applications. This makes it harder to monitor security risks, update compliance requirements, and respond quickly to possible threats.

Since law firms manage sensitive information such as client records, legal documents, financial details, and case-related information, the lack of proper security management may expose them to cyberattacks, unauthorized access, and data breaches. There is also a need for a centralized platform that can help law firms monitor security activities, manage compliance requirements, and make better decisions regarding data protection.

This study aims to address the following problems:

1. Existing law firms do not have a single platform that combines cybersecurity protection and legal compliance management in one system.
2. Current cybersecurity solutions mainly focus on protecting computer systems and networks but may not provide features that help law firms monitor legal and regulatory compliance requirements.
3. Confidential client information, including personal details, financial records, and legal documents, remains at risk due to possible cyber threats such as unauthorized access, malware attacks, and data breaches.
4. Law firms experience difficulties in tracking, updating, and maintaining compliance with the Data Privacy Act of 2012 and other related data protection requirements.
5. Existing systems lack a centralized platform that provides real-time monitoring, security reports, risk assessment, and decision-support tools for effective cybersecurity and compliance management.

1.3 Objectives of the Study

1.3.1 General Objective

The general objective of this study is to design and develop **LEXSHIELD: Integrated Cybersecurity and Legal Compliance Platform for Law Firms**. The system aims to provide law firms with a centralized platform that helps protect sensitive client information, improve cybersecurity practices, support compliance with legal requirements, and provide useful monitoring and management tools for administrators.

The proposed system is intended to help law firms manage security risks more effectively by combining cybersecurity features and compliance management processes into one platform. It will provide tools that can assist users in monitoring possible threats, organizing compliance activities, and maintaining the security of important information.

1.3.2 Specific Objectives

Specifically, this study aims to:

1. Identify and analyze the needs of law firms regarding cybersecurity and legal compliance management, including the problems and limitations of their current systems.
2. Design a secure, simple, and user-friendly interface that allows law firm users and administrators to easily access and manage the platform's features.
3. Develop integrated modules that support cybersecurity monitoring, data privacy management, and compliance tracking based on applicable regulations.
4. Implement monitoring features that can help detect possible security risks, track system activities, and provide updates regarding cybersecurity and compliance status.

5. Develop an administrative dashboard that allows authorized users to manage system activities, generate reports, review security information, and support decision-making.
 6. Create a secure and centralized database for storing and managing important client and system information while maintaining data protection.
 7. Test and evaluate the developed platform based on security, usability, performance, reliability, and effectiveness in supporting cybersecurity and compliance management.
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1.4 Scope and Delimitation

This study focuses on the design and development of **LEXSHIELD: Integrated Cybersecurity and Legal Compliance Platform for Law Firms**. The system is created specifically to help law firms improve their management of cybersecurity and legal compliance through a centralized platform.

The proposed system will include important features such as user registration, user authentication, role-based access control, secure data storage, data encryption, compliance monitoring, security risk monitoring, threat detection, and an administrative dashboard. These features are intended to help law firms protect confidential information and manage cybersecurity-related activities more efficiently.

The compliance module of the system will focus on supporting requirements related to the Data Privacy Act of 2012 and other relevant data protection practices. The platform will assist users in organizing compliance information, monitoring security activities, and generating reports that may help administrators review the status of their security and compliance efforts.

However, this study has several limitations. The system will only be developed as a prototype for academic purposes and will not be fully deployed across all law firms in the Philippines as an example. It will not include complete integration with existing law firm software, external government systems, or third-party applications. The study will also not cover advanced artificial intelligence-based prediction systems, automated legal decision-making, or large-scale cybersecurity operations.

The effectiveness of the system will be evaluated based on testing results related to usability, functionality, security features, and overall performance within the scope of the developed prototype.

1.5 Significance of the Study

The proposed **LEXSHIELD: Integrated Cybersecurity and Legal Compliance Platform for Law Firms** may provide benefits to the following groups:

- **Law Firms**
 - This study may help law firms improve the protection of confidential client information by providing a centralized platform for cybersecurity monitoring and compliance management. It may reduce the difficulty of managing different security tools and help firms organize their data protection processes more effectively.

- **Administrators and IT Personnel**

- The system may assist administrators and IT personnel by providing tools for monitoring security activities, managing user access, reviewing reports, and identifying possible risks. It may also help them maintain better control over cybersecurity and compliance tasks.

- **Clients**

- Clients may benefit from improved protection of their personal and legal information. By using a platform focused on cybersecurity and compliance, law firms can strengthen their ability to protect client data and maintain client trust.

- **BSIT Students**

- This study may serve as a learning resource for BSIT students who are interested in software development, cybersecurity, database management, and system integration. It may provide an example of applying information technology concepts to solve real-world problems.

- **Future Researchers**

- Future researchers may use this study as a reference for developing similar systems related to cybersecurity, legal technology, data privacy, and compliance management. It may also provide ideas for improving or expanding integrated security platforms in different industries.

1.6 Definition of Terms

- **Cybersecurity**

- The process of protecting computers, networks, applications, and digital information from cyber threats, unauthorized access, and possible damage.
- **Legal Compliance**
 - The process of following required laws, regulations, policies, and standards that apply to an organization's activities and operations.
- **Data Privacy Act of 2012**
 - A Philippine law that provides rules and guidelines for the proper collection, processing, storage, and protection of personal information (Republic Act No. 10173, 2012).
- **Integrated Platform**
 - A software system that combines different features and functions into one centralized application instead of using separate systems.
- **Administrative Dashboard**
 - A system interface that allows administrators to monitor activities, manage users, review information, and generate reports.
- **Real-Time Monitoring**
 - The continuous process of observing system activities, security events, and possible threats as they happen.
- **Threat Detection**
 - The process of identifying possible cybersecurity risks, attacks, or suspicious activities within a system.
- **Data Encryption**
 - A security method that converts information into a protected format to prevent unauthorized users from accessing it.
- **Role-Based Access Control**

- A security method that limits user access based on their assigned roles and responsibilities within the system.
 - **Compliance Monitoring**
 - The process of checking and tracking whether an organization follows required laws, policies, and security standards.
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CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

2.0 Introduction

This chapter presents the foreign and local studies, related literature, and system concepts that are relevant to the development of **LEXSHIELD: Integrated Cybersecurity and Legal Compliance Platform for Law Firms**. The main purpose of this chapter is to provide a strong foundation for the study by reviewing existing research, published articles, books, and other reliable sources related to cybersecurity, legal compliance, data privacy, and software development. These references help explain the current practices, challenges, and solutions that are connected to the proposed system.

Reviewing previous studies is important because it helps the researchers understand what has already been done and identify areas that still need improvement. It also provides ideas on the features, technologies, and methods that can be applied in developing the proposed platform. By examining different studies and literature, the researchers can compare various approaches and determine which ones are most suitable for the needs of law firms.

This chapter is divided into four main sections: foreign studies, local studies, related literature, and the synthesis of the reviewed works. These include topics such as cybersecurity, data privacy, information security, legal compliance, and software development. This section also discusses existing systems and tools that are related to the study and identifies their strengths and limitations. And, since law firms manage confidential client information and important legal documents, having an integrated system can help improve data security, support

compliance with legal requirements, reduce security risks, and make daily operations more efficient.

2.1 Foreign Studies

Foreign studies present ideas and findings from research done in other countries. These studies help explain how organizations handle cybersecurity, protect information, follow security rules, maintain privacy, and manage risks. They also provide useful examples, tools, and practices that can serve as references in developing the proposed cybersecurity compliance management system for law firms. For easier presentation and discussion, the foreign studies are organized into the following categories.

A. Cybersecurity Challenges and Protection Needs in Law Firms

“Law Firm Cybersecurity: The State of Preventative and Remedial Regulation Governing Data Breaches in the Legal Profession”

Tarr (2017)

Tarr (2017) explored the growing cybersecurity challenges faced by law firms and explained why legal professionals play an important role in protecting confidential client information. The paper described how law firms handle highly sensitive records, such as legal documents, privileged communications, financial information, and personal data, making them attractive targets for cybercriminals. It discussed how data breaches can affect client confidentiality, damage a firm's reputation, and lead to legal and financial consequences. The author emphasized that these risks can be reduced by adopting preventive cybersecurity measures, complying with applicable regulations, and preparing effective incident response plans. The study concluded that law firms should establish clear cybersecurity policies and strengthen their security practices to safeguard client information and maintain public trust in legal services.

This study is closely related to LEXSHIELD because it shows the importance of protecting confidential information handled by law firms. Its discussion on preventive security measures, regulations, and incident response supports the cybersecurity and compliance features included in LEXSHIELD.

“Cybersecurity in Accounting Research: Current State and Future Directions”

Haapamäki & Sihvonen (2019)

They examine the current condition of cybersecurity research in the accounting field. Their study reviewed 39 theoretical and empirical publications to identify the main topics that researchers have explored over the years. These topics included cybersecurity information sharing, cybersecurity investments, internal auditing and security controls, cybersecurity disclosures, and security threats and data breaches. The authors explained that cybersecurity is not limited to using advanced technologies or security software. Instead, organizations also need effective internal controls, proper risk management strategies, clear security policies, and active participation from management and employees to protect sensitive information. The review also showed that cyber threats continue to become more complex, creating new challenges for organizations that handle confidential financial and client data. Because of these changing risks, the authors emphasized the need for more research to develop better cybersecurity practices, improve organizational preparedness, and support stronger decision-making in cybersecurity management. Overall, the study provides a clear overview of existing cybersecurity research and offers a useful framework that researchers and organizations can use to strengthen cybersecurity programs, improve security governance, and better protect valuable information from cyber threats.

The study provides a broader view of cybersecurity by showing that protecting information requires more than security software. Its focus on internal controls, risk

management, security policies, and employee involvement supports LEXSHIELD's goal of combining cybersecurity practices with proper compliance management.

"Does Cybersecurity Regulation Reduce Corporate Data-Breach Risk?"

Wang et al. (2025)

They investigated the effectiveness of cybersecurity regulations in reducing corporate data-breach risks by analyzing data from 3,761 publicly listed firms in China. The researchers examined how regulatory requirements influence organizations' cybersecurity practices and whether compliance with these regulations contributes to stronger protection against cyber incidents. The study focused on understanding the relationship between government-imposed cybersecurity policies and organizational efforts to safeguard sensitive information.

The findings showed that cybersecurity regulations significantly reduce the likelihood of corporate data breaches by encouraging organizations to strengthen their cybersecurity culture, enhance security governance, and increase investments in cybersecurity technologies and protective measures. The researchers also found that organizations subject to stronger regulatory oversight were more likely to implement effective security controls and improve their overall preparedness for cyber threats. The study concluded that cybersecurity regulations are most effective when organizations integrate compliance requirements into their overall cybersecurity strategy rather than treating them merely as legal obligations. These results demonstrate that legal compliance, sound governance, and proactive cybersecurity management work together to improve organizational resilience and protect confidential information from evolving cyber risks.

The findings show that following cybersecurity regulations can help organizations reduce the risk of data breaches. This supports LEXSHIELD's approach of connecting cybersecurity

measures with legal compliance so that law firms can better protect sensitive information while meeting security requirements.

"Client Confidentiality as Data Security"

Perlin (2024)

With the increasing reliance of law firms on digital technologies, Perlin (2024) examined how the concept of attorney-client confidentiality has expanded beyond traditional methods of protecting information. The study explained that lawyers are now responsible not only for preventing unauthorized disclosure of client information but also for ensuring that digital platforms, cloud services, and electronic communication systems are secured against possible cyber threats.

The research emphasized that law firms handle highly sensitive information, including legal documents, case records, confidential discussions, and personal data, which makes cybersecurity an important part of protecting client relationships and trust. Perlin discussed how the use of modern technologies introduces new security challenges, such as unauthorized access, data exposure, and vulnerabilities in digital storage systems. The author highlighted the need for law firms to implement effective information security practices, make responsible technology choices, and establish procedures that minimize the impact of cybersecurity incidents. The study concluded that maintaining client confidentiality in today's digital environment requires lawyers to view cybersecurity as a core professional obligation and not merely as a technical issue managed by information technology personnel.

This study directly supports LEXSHIELD because it explains that protecting client confidentiality now also requires securing digital systems and electronic information. It provides a basis for including security measures in LEXSHIELD to help law firms protect case records, legal documents, and other confidential client information.

"When Should Law Firms Notify Clients About Data Breaches?"

Goldsmith, Standish, & Temkin (2021)

Goldsmith, Standish, and Temkin (2021) examined the challenges faced by law firms in responding to cybersecurity incidents involving the exposure of confidential client information. The study discussed how legal organizations have become attractive targets for cybercriminals because they handle valuable data, including client records, case documents, litigation files, and private communications. The authors highlighted that data breaches, ransomware attacks, and unauthorized access incidents can create serious risks for both law firms and their clients.

The research focused on the responsibility of law firms to determine when clients should be informed after a cybersecurity incident occurs. The authors explained that firms must assess factors such as the severity of the breach, the type of information compromised, and the potential effects on affected clients before deciding on proper notification actions. They emphasized the importance of having established cybersecurity policies, incident response plans, and communication procedures to manage security incidents effectively.

The study concluded that cybersecurity readiness is an essential part of modern legal practice. Proper handling of data breaches helps law firms protect client confidentiality, maintain professional obligations, preserve client trust, and reduce the negative effects of cyberattacks on their operations and reputation.

This research highlights the need for law firms to have clear procedures when a data breach occurs, especially when client information is affected. Its discussion of incident response, breach assessment, and client notification supports LEXSHIELD's security monitoring and reporting features for handling cybersecurity incidents.

B. Information Security Governance, Privacy, and Cybersecurity Frameworks

“Privacy by Design: The 7 Foundational Principles”

Cavoukian (2011)

The author emphasized that protecting personal information should begin long before an information system is completed. Through the concept of Privacy by Design, Cavoukian (2011) explained that privacy should be included in every stage of system planning, design, and development instead of being treated as an additional feature after problems arise. The paper described seven principles that guide organizations in creating systems that protect users' personal information. These principles include making privacy the default setting, ensuring transparency in data handling, implementing effective security safeguards, and giving users greater control over their information. By following these principles, organizations can reduce privacy risks, improve the security of their systems, and strengthen the confidence of users in how their personal data is managed.

Cavoukian's study supports LEXSHIELD's focus on including privacy and security from the beginning of system development. The principles presented in the study provide a basis for designing LEXSHIELD in a way that protects personal and confidential information while giving importance to privacy throughout the system.

“The Need for Information Security Governance: An Enterprise Perspective”

Posthumus & von Solms (2004)

According to Posthumus and von Solms (2004), organizations need more than technical security tools to effectively protect their information and manage today's cybersecurity challenges. The paper explained that technologies such as firewalls, antivirus software, and encryption are important, but they cannot provide complete protection when used on their own. The authors argued that organizations should also establish formal information security

governance that includes clear policies, well-defined procedures, effective risk management, and accountability at every level. These governance practices help ensure that security measures are consistently applied and supported by both management and employees. The paper also pointed out that having a structured governance framework allows organizations to protect valuable information assets while meeting business objectives and complying with legal and regulatory requirements. Overall, the authors emphasized that combining technical security with strong governance creates a more effective and organized approach to managing cybersecurity risks and protecting sensitive information.

The study supports the need for a structured approach to managing information security through policies, procedures, risk management, and accountability. These ideas are reflected in LEXSHIELD through its security and compliance features, which are intended to help law firms protect sensitive information in an organized way.

***"Information Security Governance Challenges and Critical Success Factors:
Systematic Review"***

Alshaikh (2020)

This systematic review examined the different challenges organizations face when implementing information security governance. The researchers analyzed previous studies to identify the most common issues that affect the successful management of information security. The review found that many organizations struggle with implementing security policies consistently, managing cybersecurity risks effectively, aligning security practices with legal and regulatory compliance requirements, and maintaining continuous monitoring of their information systems. It also highlighted that these challenges are often influenced by organizational factors such as limited resources, lack of management support, insufficient employee awareness, and rapidly changing cyber threats. Based on the findings, the study concluded that effective cybersecurity governance cannot rely on technology alone. Instead, it requires well-defined

governance frameworks, strong commitment from organizational leaders, clear policies and responsibilities, regular risk assessments, continuous monitoring, and ongoing evaluation to ensure that security measures remain effective and responsive to emerging threats. Overall, the review emphasizes that a structured and proactive approach to information security governance is essential for protecting organizational information assets and improving an organization's overall cybersecurity posture.

Alshaikh's findings are relevant to LEXSHIELD because they identify common problems such as weak policies, poor risk management, and lack of continuous monitoring. These concerns helped support the inclusion of monitoring, security management, and compliance-related functions in the proposed platform.

"Cybersecurity Frameworks and Their Role in Improving Organizational Security"

Calder & Watkins (2019)

The authors explained that cybersecurity frameworks provide organizations with a structured method for improving their security management practices. Calder and Watkins (2019) discussed how standards such as ISO/IEC 27001 help organizations identify possible security risks, establish appropriate controls, monitor the effectiveness of security measures, and prepare for regulatory compliance. The paper emphasized that organizations with well-defined cybersecurity frameworks are better equipped to manage security challenges and respond to changing threats. It also noted that these frameworks encourage continuous assessment and improvement, allowing organizations to strengthen their overall security posture over time. The authors concluded that adopting standardized cybersecurity frameworks supports consistent security practices, improves organizational readiness, and enhances the protection of valuable information assets.

This study provides support for using established cybersecurity frameworks when developing security practices. Its focus on risk identification, security controls, monitoring, and continuous improvement is useful in shaping the cybersecurity functions of LEXSHIELD.

"Assessing Frameworks for Eliciting Privacy & Security Requirements from Laws and Regulations"

Oluwafemi Olukoya (2022)

The study evaluates different frameworks used to identify privacy and security requirements from laws and regulations for software development. As organizations collect and manage more personal data, complying with privacy laws has become increasingly important. However, converting legal requirements into software specifications remains challenging because legal documents are often complex and difficult to interpret.

The research reviews existing requirement elicitation frameworks and examines whether they can be applied beyond commonly studied regulations such as the GDPR and HIPAA. To assess their effectiveness, the study uses the Nigerian Data Protection Regulation (NDPR) as a case study and compares the extracted requirements with the privacy practices of FinTech organizations.

The findings show that current frameworks are useful for identifying legal privacy and security requirements but have limitations when applied to different legal systems. The study recommends developing more flexible and standardized frameworks that can help software engineers create systems that are both secure and compliant with various privacy laws.

The study is important to LEXSHIELD because it shows the difficulty of turning legal requirements into actual software requirements. Its findings support the need for LEXSHIELD to

connect privacy and security requirements with system features so that law firms can manage compliance more effectively.

"Arguing Regulatory Compliance of Software Requirements"

G. Siena, A. Perini, A. Susi, & J. Mylopoulos (2013)

The study introduces a framework that helps software developers determine whether software requirements comply with legal and regulatory obligations during the software development process. It uses argumentation techniques to establish clear relationships between legal regulations and software requirements, allowing developers to justify why specific system features satisfy legal provisions. This structured approach makes compliance analysis more transparent, organized, and easier to verify.

Instead of treating regulatory compliance as a task performed after software has been developed, the framework integrates legal requirements into the early stages of system design. By identifying possible compliance issues during requirements engineering, developers can resolve legal conflicts before implementation, reducing the need for costly changes later in the development process. The framework also supports documenting compliance decisions, making it easier for organizations to demonstrate that their software follows applicable laws and regulations.

Overall, the study shows that incorporating legal compliance into requirements engineering improves software quality, minimizes legal and regulatory risks, and helps organizations develop systems that are both functional and compliant with relevant legal standards.

This research supports the idea of considering legal requirements during the early stages of software development rather than only checking compliance after the system is finished. This

approach is useful for LEXSHIELD because legal and privacy requirements are part of the system's design and compliance functions.

"An Approach Based on Citation Analysis to Support Effective Handling of Regulatory Compliance"

M. Ghanavati, D. Amyot, & L. Peyton (2011)

This research study presents a method that helps organizations manage and understand legal and regulatory requirements more effectively by using citation analysis. Since laws and regulations often refer to other legal documents, organizations may find it difficult to identify how different rules are connected and how changes in one regulation can affect others. The proposed approach analyzes these legal references to create clear links between related regulations, making them easier to organize, understand, and interpret.

The method also helps track updates and changes in legal documents, allowing organizations to identify which compliance requirements may be affected and what actions may be needed to remain compliant. It provides a structured way to examine legal relationships, making it easier to monitor regulatory changes and assess their impact on organizational policies and software requirements. This approach can also improve communication between legal experts and software developers by presenting regulatory information in a more organized and understandable way.

Thus, the study shows that using citation analysis can improve the management of regulatory compliance, reduce the risk of missing important legal obligations, and help organizations maintain compliance as laws and regulations continue to evolve.

The study shows how organizing relationships between laws and regulations can make compliance requirements easier to understand and manage. This is relevant to LEXSHIELD

because the platform can help law firms organize and monitor their compliance requirements instead of handling them manually.

"Governing the Fiduciary Relationship in Information Security Services"

Kankanhalli, Teo, Tan, & Wei (2016)

According to Kankanhalli, Teo, Tan, and Wei (2016) examined how governance mechanisms influence the relationship between organizations and information security service providers. The study focused on the fiduciary nature of information security services, emphasizing that organizations must establish appropriate governance structures to ensure that service providers act in the best interests of their clients. The authors explained that effective governance promotes accountability, clearly defines roles and responsibilities, and builds trust between organizations and external security providers.

The researchers further discussed that successful information security governance extends beyond technical safeguards by incorporating managerial oversight, contractual controls, and continuous collaboration between stakeholders. Their findings showed that organizations with well-defined governance mechanisms are better able to align cybersecurity initiatives with business objectives, improve decision-making, and enhance the protection of organizational information assets. The study concluded that strong governance frameworks contribute to higher information security performance by fostering accountability, supporting compliance efforts, and ensuring that security responsibilities are effectively managed across the organization.

This research emphasizes accountability, clear responsibilities, and proper governance in information security. These concepts are useful to LEXSHIELD because the platform includes user management, activity monitoring, and security controls that can help law firms keep track of security-related responsibilities.

"Policy Development and Frameworks for Cybersecurity in Corporates and Law Firms"

Bou Alwan (2018)

According to Bou Alwan (2018), although organizations continue to improve their cybersecurity practices and invest in better ways to reduce security risks, cyberattacks are still increasing at a rapid rate. Advances in technology have made communication, business operations, and information sharing faster and more convenient, but they have also created more opportunities for cybercriminals to attack organizations. The author highlights reports showing a significant rise in malware and ransomware incidents, demonstrating that cyber threats have become more frequent and more sophisticated over time. Because of these growing risks, corporations and law firms need to establish clear cybersecurity policies and adopt effective security frameworks to protect confidential information, reduce vulnerabilities, and prepare for possible cyber incidents.

The study emphasizes that having well-developed cybersecurity policies, supported by continuous monitoring and regular updates, is essential for protecting organizational data, maintaining business operations, and ensuring that legal and corporate organizations can respond effectively to evolving cyber threats.

Bou Alwan's study is directly related to LEXSHIELD because it discusses cybersecurity policies and frameworks for law firms and other organizations. Its findings support the need for clear security policies, regular monitoring, and updated protection measures to help law firms deal with changing cyber threats.

"Cybersecurity Laws and Ethical Implications: Integrated Cybersecurity Governance Model (CLEI-ICGM)"

Ababneh et al. (2026)

The study explains the growing need for a stronger cybersecurity governance framework that combines legal compliance, ethical responsibility, and technical security. The authors point out that many existing cybersecurity approaches treat legal, ethical, and technical concerns separately, which often results in inconsistent policies and weak security practices. To address this problem, they developed the Integrated Cybersecurity Governance Model (ICGM), a framework that brings these three areas together into one unified approach. The study reviewed 50 peer-reviewed publications published between 2020 and 2025 to identify gaps in cybersecurity governance and examine new ethical and regulatory challenges brought by artificial intelligence. The proposed model was also evaluated through a survey involving 500 participants from academia, industry, and government, with statistical tests confirming its reliability and effectiveness.

The findings show that the ICGM helps organizations make better cybersecurity decisions by integrating AI ethics, improving compliance with cybersecurity laws, strengthening proactive threat management, and increasing operational resilience. The study concludes that adopting this integrated governance model can help organizations build more secure, ethical, and legally compliant digital environments while supporting future improvements in cybersecurity policies and digital infrastructure.

The study supports LEXSHIELD's integrated approach because it combines cybersecurity, legal compliance, and ethical considerations instead of treating them as separate concerns. This is similar to LEXSHIELD's goal of bringing cybersecurity and legal compliance functions together in one platform.

**"Cybersecurity Laws, Digital Crimes, and Ethical Considerations: A
Multidimensional Perspective on Cyber Risk Regulation (MPoCRR)"
Ababneh et al. (2026)**

Cybersecurity is no longer only a technical concern because it now involves legal, ethical, and governance issues as digital technologies such as cloud computing, artificial intelligence (AI), and the Internet of Things (IoT) continue to grow. The authors compared major international cybersecurity and data protection regulations, including the General Data Protection Regulation (GDPR), the Health Insurance Portability and Accountability Act (HIPAA), and the Network and Information Security (NIS) Directive, to examine how different countries address cyber risks and digital crimes. Using a qualitative approach and information gathered from secondary sources, the study found that countries differ in how they define cybercrime, enforce cybersecurity laws, and include ethical principles in their policies. The findings also show that current cybersecurity regulations are often fragmented and reactive, making them less effective in responding to modern threats such as AI-driven surveillance and algorithmic bias.

To address these challenges, the researchers proposed the Multidimensional Perspective on Cyber Risk Regulation (MPoCRR), which combines ethical AI principles, legal compliance, and data protection into a unified cybersecurity governance framework. The study concludes that adopting this model can help organizations and governments develop more flexible, transparent, and ethically guided cybersecurity policies that are better prepared to address emerging digital threats.

This research highlights the need to consider legal compliance, data protection, and ethical issues when managing modern cyber risks. Its proposed integrated approach is relevant to LEXSHIELD because the system is designed to help law firms address cybersecurity and compliance concerns through a centralized platform.

"The EU Cyber Resilience Act: Hybrid Governance, Compliance, and Cybersecurity Regulation in the Digital Ecosystem"

Teichmann and Sergi (2025)

According to Teichmann and Sergi (2025), the European Union Cyber Resilience Act (CRA) introduces a new approach to cybersecurity governance by combining government regulations with the responsibility of private organizations to manage cybersecurity risks. The authors explain that the law requires manufacturers and businesses to follow cybersecurity requirements throughout the entire lifecycle of digital products, from development and production to maintenance and security updates. They also discuss how the CRA works together with other European cybersecurity regulations, including the NIS2 Directive and the Cybersecurity Act, to create a more coordinated system for improving digital security. In addition, the authors compare the European approach with cybersecurity policies in the United States and the United Kingdom to show the differences in how these countries regulate cybersecurity and compliance.

They conclude that the CRA strengthens cybersecurity by clearly defining the responsibilities of both public authorities and private organizations, promoting the use of harmonized standards, improving regulatory compliance, strengthening market oversight, and helping organizations prepare for future cybersecurity challenges through continuous security improvements.

The study demonstrates how cybersecurity regulations can require organizations to continuously manage security risks and maintain compliance. This supports LEXSHIELD's compliance monitoring approach, particularly its goal of helping law firms track security requirements and improve their cybersecurity practices over time.

C. Cybersecurity Compliance Platforms and Integrated Security Systems

" A Customizable Web Platform to Manage Standards Compliance of Information Security and Cybersecurity Auditing "

Antunes, Maximiano, & Gomes (2022)

This study developed a customizable web-based platform designed to improve the management of cybersecurity auditing and compliance requirements within organizations. The researchers explained that traditional compliance processes are often fragmented, time-consuming, and heavily dependent on manual documentation, making it difficult for organizations to monitor compliance activities and maintain accurate records. These challenges can lead to delays in audits, inconsistent reporting, and difficulty in identifying security issues. To address these problems, the proposed platform integrates several important functions into a single system, including security auditing, compliance evaluation, documentation management, and corrective action tracking. By centralizing these processes, the platform enables organizations to organize compliance information more efficiently, monitor audit progress, and respond more quickly to identified security gaps.

The results show that using a centralized web-based platform improves the efficiency of cybersecurity audits, reduces the administrative burden associated with manual processes, and helps organizations detect security weaknesses more effectively. Overall, the study demonstrates that integrated compliance management systems can strengthen cybersecurity governance, support regulatory compliance, and improve an organization's ability to maintain a secure and well-managed information security environment.

This study is highly relevant to LEXSHIELD because it shows how a web-based platform can bring cybersecurity auditing and compliance activities into one system. Its use of centralized

records, compliance monitoring, and corrective action tracking supports the development of similar functions in LEXSHIELD for law firms.

" Cyber Security and Compliance Management Through a Single Integrated Platform "

Jagtap, Pagar, & Meshram (2018)

According to Jagtap, Pagar, and Meshram (2018), organizations can improve their cybersecurity by using a single platform that combines different security functions into one system. The paper presented an integrated cybersecurity platform that includes security assessment, vulnerability analysis, auditing, and compliance management. The authors explained that many organizations rely on separate security tools to perform these tasks, which often leads to fragmented monitoring, duplicated work, and delays in responding to security issues. They pointed out that managing multiple systems can make cybersecurity operations more difficult and less efficient. By bringing these functions together in one centralized platform, organizations can monitor their security environment more effectively, simplify compliance activities, and improve their ability to detect and respond to potential threats. Overall, the paper emphasized that an integrated platform helps strengthen cybersecurity management while making security operations more organized and efficient.

The study supports the main concept of LEXSHIELD by showing the benefits of combining cybersecurity assessment, vulnerability analysis, auditing, and compliance management in one platform. This approach can help LEXSHIELD reduce the need for separate tools and make security monitoring and compliance activities easier for law firms to manage.

" Compliance Management Systems: Do They Make a Difference? "

Coglianesse & Nash (2021)

The research emphasized that compliance management systems are valuable tools for helping organizations meet legal and regulatory requirements in a more organized and efficient manner. Coglianesse and Nash (2021) explained that these systems assist organizations in identifying compliance obligations, assigning responsibilities, monitoring performance, and correcting problems before they become more serious. The paper pointed out that technology improves the efficiency of compliance activities by making information easier to manage and helping organizations monitor their progress more effectively. At the same time, the authors stressed that successful compliance depends on strong governance, clear policies, and active support from organizational leaders. Overall, the paper showed that combining technology with proper governance strengthens accountability, improves risk management, and helps organizations maintain consistent compliance with applicable laws and standards.

The research shows how compliance management systems can help organizations identify requirements, monitor compliance, assign responsibilities, and address problems. These findings provide support for LEXSHIELD's compliance features, particularly those intended to help law firms organize their legal and cybersecurity obligations.

" Holistic Information Security Management and Compliance Framework "

Grigaliūnas et al. (2024)

The authors emphasized the importance of adopting a holistic approach to information security management by combining cybersecurity controls, governance practices, and compliance requirements into a single framework. Grigaliūnas et al. (2024) explained that many organizations manage these areas separately, which can create gaps in security and increase the risk of cyber threats and regulatory issues. The paper pointed out that an integrated framework helps organizations coordinate their security activities more effectively by ensuring that technical controls, organizational policies, and compliance efforts work together.

The authors also noted that this approach improves communication among different departments, supports better decision-making, and strengthens the organization's ability to manage security risks. Overall, the paper concluded that a unified information security framework helps organizations improve regulatory compliance, protect sensitive information, and establish a more consistent and effective cybersecurity management system.

This study supports the idea of connecting cybersecurity, governance, and compliance instead of managing them separately. Its findings are useful for LEXSHIELD because the platform is designed to bring security controls and compliance activities together to provide a more organized way of protecting sensitive legal information.

“Applying Information Security Governance”

Moulton & Coles (2003)

The study by Moulton and Coles (2003) explored the role of information security governance as an essential component of effective organizational management. As businesses became increasingly dependent on information systems and faced stricter corporate governance requirements, the authors argued that information security should no longer be viewed solely as a technical concern but as a strategic management responsibility. They explained that organizations need a structured governance framework that clearly defines security policies, roles, responsibilities, and accountability to effectively manage information security risks.

In addition, the study emphasized that successful information security governance requires the integration of risk management, internal controls, and continuous monitoring into an organization's overall governance structure. The authors noted that senior management plays a critical role in establishing security objectives, allocating resources, and ensuring compliance with organizational policies and regulatory requirements. By embedding information security into business decision-making processes, organizations can strengthen the protection of their

information assets, improve operational efficiency, reduce security vulnerabilities, and enhance stakeholder confidence. The study concluded that a governance-based approach provides organizations with a more sustainable and effective way to manage information security than relying solely on technical security solutions.

Moulton and Coles emphasize that information security should be part of organizational management and not depend only on technical tools. This is relevant to LEXSHIELD because the platform includes management, monitoring, policies, and security functions that can help law firms handle information security in a more organized manner.

"Fortifying Digital Justice: A Cybersecurity and Efficiency Framework for U.S. Legal SMEs and Court-Affiliated Service Providers"

Isabirye (2025)

The study discusses how the growing use of digital technology in the legal industry has created many opportunities to improve legal services, but it has also increased the risk of cyberattacks and data security issues. Small and medium-sized legal enterprises (SMEs) and court-affiliated service providers in the United States are particularly affected because many of them still depend on outdated and manual processes while dealing with more advanced cybersecurity threats. To help solve these challenges, the researcher introduced a framework that combines strong cybersecurity measures with strategies for improving operational efficiency. The proposed framework includes key practices such as conducting regular risk assessments, managing user identities and access, encrypting sensitive information, automating routine workflows, and adopting cloud-based technologies. These components work together to strengthen security while making legal services more organized and efficient.

The study concludes that implementing this integrated framework can help legal organizations better protect confidential client information, improve their daily operations, comply

with legal and regulatory requirements, and provide faster, safer, and more accessible legal services to the public.

This study is particularly relevant because it focuses on cybersecurity and efficiency in legal organizations. Its recommendations on risk assessment, access control, encryption, and workflow improvement provide useful support for the security and management features included in LEXSHIELD.

D. Data Privacy, Regulatory Compliance, and Legal Information Protection

" The EU General Data Protection Regulation (GDPR): A Practical Guide "

Paul Voigt and Axel von dem Bussche (2017)

This research examined the requirements and implications of the General Data Protection Regulation (GDPR) for organizations handling personal information. The authors explained that organizations must implement appropriate technical and organizational security measures to protect personal data and demonstrate compliance with privacy regulations. The study emphasized accountability, documentation, data protection policies, and continuous compliance monitoring as essential practices for organizations managing sensitive information.

This study is relevant to LEXSHIELD because it explains the importance of protecting personal data through proper security measures, documentation, and accountability. These practices support the compliance functions of LEXSHIELD, particularly in helping law firms manage sensitive client information and monitor their data protection responsibilities.

" Cybersecurity Threats and Vulnerabilities: A Systematic Review "

Al-Rimy, Maarof, & Shaid (2019)

This systematic review examined the different cybersecurity threats, weaknesses, and attack techniques that affect organizations in the digital environment. The researchers found that

organizations continue to experience growing security risks caused by threats such as malware, phishing attacks, unauthorized access, and data breaches. The study highlighted that relying only on reactive measures is not enough to protect sensitive information. Instead, organizations should implement proactive security practices, including continuous monitoring, regular vulnerability assessments, strong security controls, and effective risk management strategies. The research concluded that adopting a comprehensive cybersecurity approach is necessary to improve information protection and reduce the impact of evolving cyber threats.

The study provides a basis for the security monitoring and risk management features of LEXSHIELD. Its emphasis on continuous monitoring, vulnerability assessment, and proactive security measures supports the system's goal of helping law firms identify and respond to possible cybersecurity risks.

" Blockchain's Roles in Strengthening Cybersecurity and Protecting Privacy "

Nir Kshetri (2017)

The research highlighted the growing importance of emerging technologies in improving cybersecurity and protecting sensitive information. Kshetri (2017) explained that organizations can strengthen their security by adopting secure information management approaches that help maintain data integrity, improve authentication, and reduce the risk of unauthorized access. The paper pointed out that traditional security methods alone may no longer be enough as cyber threats continue to become more advanced. For this reason, organizations are encouraged to implement stronger and more innovative security mechanisms that can better protect digital information. The author also emphasized that improving cybersecurity is essential for maintaining trust, safeguarding privacy, and ensuring the secure use of digital technologies. By adopting these approaches, organizations can reduce security risks and improve the overall protection of their information systems.

Kshetri's research shows the importance of using stronger security approaches to protect digital information and maintain data privacy. Although LEXSHIELD does not necessarily depend on blockchain technology, the study supports the system's focus on data protection, secure access, and reducing the risk of unauthorized use of sensitive information.

“Legal Policies Failing on Data Breaches? An Empirical Study of U.S. Information Security Law Implementations”

Zhang et al. (2023)

Zhang et al. (2023) investigated whether information security laws in the United States have been effective in reducing the occurrence of data breaches. Recognizing that governments have introduced and revised numerous legal policies to strengthen data protection, the researchers examined panel data on reported data breaches in the United States from 2005 to 2019 to assess whether these regulations have achieved their intended purpose. The study also evaluated the effectiveness of legal penalties in discouraging organizations from neglecting information security responsibilities.

Furthermore, the researchers sought to determine whether existing legal frameworks serve as effective deterrents against cybersecurity incidents or whether additional policy improvements are necessary. By analyzing the relationship between legal enforcement and the frequency of data breaches, the study provides empirical evidence on the strengths and limitations of current information security laws. The findings offer valuable insights for policymakers and organizations in developing more effective cybersecurity regulations, strengthening compliance mechanisms, and enhancing the protection of sensitive information against evolving cyber threats.

This study shows that having information security laws alone does not guarantee that organizations will be protected from data breaches. Its findings support LEXSHIELD's approach

of combining legal compliance with actual cybersecurity measures to help law firms put security requirements into practice.

"Challenges and Enablers for GDPR Compliance: Systematic Literature Review and Future Research Directions"

Zaguir, Magalhaes, & Spinola (2024)

According to Zaguir, Magalhaes, and Spinola (2024), organizations face several challenges when trying to comply with the General Data Protection Regulation (GDPR). Through a systematic review of previous studies, the researchers identified common issues such as the need to change organizational processes, limited technological capabilities, lack of knowledge about compliance requirements, and difficulties in turning legal obligations into practical security measures. These challenges make it difficult for many organizations to fully implement GDPR requirements.

The study also explains that GDPR compliance is not only about following legal rules but also about building strong privacy practices within the organization. This includes improving information governance, applying appropriate security controls, and continuously monitoring how personal data is managed. The researchers found that organizations are more successful when they use structured compliance frameworks, enterprise architecture management, and effective information governance to support their compliance efforts.

Thus, the findings show that combining legal compliance with cybersecurity strategies helps organizations strengthen data protection and reduce privacy risks. The researchers emphasize the value of integrated compliance solutions that assist organizations in managing privacy obligations, monitoring security practices, and maintaining accountability.

The study identifies difficulties organizations face when turning privacy laws into practical security and organizational practices. This is useful for LEXSHIELD because the platform aims to make compliance easier to manage by connecting privacy requirements, security practices, monitoring, and documentation in one system.

"Are We There Yet? Understanding the Challenges Faced in Complying with the General Data Protection Regulation (GDPR)"

Sirur, Nurse, & Webb (2018)

Sirur, Nurse, and Webb (2018) explained that many organizations experience practical difficulties while trying to comply with the General Data Protection Regulation (GDPR). Their study examined how organizations understand GDPR requirements, make the necessary operational changes, and determine whether their data protection practices meet the required standards. The research focused on the real challenges encountered during the implementation process.

The researchers found that organizations often struggle because GDPR contains complex legal requirements that are not always easy to interpret. Many also have difficulty identifying and managing personal data, understanding their legal responsibilities, and creating effective methods to verify compliance. In addition, organizations with limited privacy knowledge and weak data management practices face greater challenges in meeting GDPR requirements.

The study concluded that successful GDPR compliance requires organizations to adopt structured approaches for managing personal information, strengthening cybersecurity practices, and maintaining clear visibility over their data processing activities. Continuous monitoring, proper documentation, and good security awareness also play important roles in maintaining compliance. These findings provide useful guidance for developing systems that support

automated compliance monitoring, privacy management, and the secure handling of confidential information.

This research highlights the difficulty of understanding and applying complex privacy requirements in actual organizations. Its findings support the need for LEXSHIELD to provide organized compliance monitoring and documentation that can help law firms keep track of their privacy and data protection responsibilities.

"The General Data Protection Regulation: Requirements, Architectures, and Constraints"

Hjerppe, Ruohonen, & Leppänen (2019)

In their study, Hjerppe, Ruohonen, and Leppänen (2019) discussed how the General Data Protection Regulation (GDPR) affects software development and system architecture. The researchers explored ways organizations can translate legal privacy requirements into technical system requirements that can be implemented during software development. Their study highlights the importance of designing information systems that support privacy and security from the beginning.

The researchers identified several key GDPR requirements that software systems should include, such as protecting personal data, providing user privacy controls, ensuring transparency, maintaining accountability, and supporting secure data processing. They also found that many organizations, especially small and medium-sized enterprises, face challenges because of limited financial resources, technical expertise, and system development capabilities needed to build privacy-compliant architectures. The findings emphasize that privacy and security should be considered during the early planning and design stages of software development instead of being added after the system has already been built. Integrating legal

requirements into the system development process helps organizations create more secure, reliable, and compliant information systems.

This study is closely connected to the development of LEXSHIELD because it explains how privacy requirements can be considered when designing software systems. Its emphasis on privacy, security, user controls, and accountability supports the inclusion of these principles during the design and development of LEXSHIELD.

"The General Data Protection Regulation and the Rise of Certification as a Regulatory Instrument"

Lachaud (2018)

Lachaud (2018) examined how certification mechanisms under the General Data Protection Regulation (GDPR) can help organizations demonstrate compliance with data protection requirements. The study analyzed how certification serves as a practical tool for showing that an organization's data processing activities follow privacy regulations and accepted security standards. It also explored how certification promotes accountability and encourages organizations to improve their privacy management practices.

The study explains that traditional regulatory approaches mainly depend on government enforcement after organizations violate the law. In contrast, certification encourages organizations to maintain compliance continuously by regularly reviewing and improving their privacy and security measures. The author also noted that certification helps organizations build trust with customers, business partners, and regulators by showing their commitment to protecting personal information responsibly.

The study revealed the importance of using compliance management tools that can document security practices, monitor privacy requirements, maintain proper records, and provide

evidence of compliance whenever necessary. The study suggests that integrating cybersecurity management with legal compliance functions enables organizations to manage their responsibilities more effectively. These findings support the development of integrated platforms that help organizations maintain both cybersecurity controls and legal compliance in an organized and efficient manner.

Lachaud's study supports the use of documentation and evidence when organizations need to show that they are following privacy requirements. This is relevant to LEXSHIELD because its compliance functions can help law firms keep records of security practices, monitor requirements, and generate information that can support their compliance activities.

E. Risk Management, Threat Detection, and Incident Response

" How Integration of Cyber Security Management and Incident Response Enables Organizational Learning "

Ahmad, Maynard, & Shanks (2020)

The paper explored how integrating cybersecurity management with incident response processes can help organizations become more resilient against cyber threats. Ahmad, Maynard, and Shanks (2020) explained that organizations are better prepared to respond to security incidents when they have a structured process for detecting, documenting, analyzing, and managing cybersecurity events. The authors pointed out that every security incident provides valuable lessons that can be used to improve future security strategies and reduce the likelihood of similar attacks. They also emphasized that incident response should not only focus on recovering from an attack but should also support continuous learning and improvement. By reviewing past incidents and applying the lessons learned, organizations can strengthen their cybersecurity capabilities, improve decision-making, and respond more effectively to future threats. Overall, the paper shows that combining cybersecurity management with incident

response creates a stronger and more proactive approach to protecting organizational information and systems.

This study supports LEXSHIELD's incident response features by showing the importance of detecting, documenting, and reviewing security incidents. Its focus on learning from previous incidents can help LEXSHIELD support continuous improvement in the security practices of law firms.

" Information Security Management Needs More Holistic Approach: A Literature Review "

Soomro, Shah, & Ahmed (2016)

The paper examined the importance of adopting a holistic approach to information security management in organizations. Soomro, Shah, and Ahmed (2016) explained that effective cybersecurity cannot be achieved by relying on technology alone. Instead, organizations need to combine technical security measures with well-defined policies, employee awareness programs, and organizational processes to build a stronger security environment. The authors pointed out that each of these components plays an important role in protecting information assets and reducing the risk of cyber threats. They also emphasized that when technology, people, and organizational practices work together, organizations are better prepared to prevent security incidents and respond effectively when they occur. The paper concluded that an integrated approach to information security management helps organizations strengthen their overall cybersecurity, improve risk management, and ensure the continuous protection of sensitive information in an increasingly digital environment.

The study shows that effective information security requires a combination of technology, policies, employee awareness, and organizational processes. This supports LEXSHIELD's

integrated approach, where cybersecurity and compliance are managed together rather than relying only on technical security tools.

"Cybersecurity Risk Management: A Review of Current Challenges and Future Directions"

Amin, T., Ahmad, A., & Maynard, S. B. (2019)

According to Amin, Ahmad, and Maynard (2019), organizations face increasing challenges in managing cybersecurity risks because digital technologies and cyber threats continue to evolve. The study reviewed different cybersecurity risk management approaches and examined how organizations identify, assess, and respond to security threats. The researchers found that traditional risk management methods are often no longer enough since cyberattacks have become more advanced, frequent, and difficult to predict.

The study explains that organizations need structured risk management frameworks that include threat identification, vulnerability assessment, security controls, continuous monitoring, and incident response planning. The authors also emphasized that effective cybersecurity risk management is not limited to using technical security tools. Organizations must also establish strong governance, develop clear security policies, and improve employee awareness to strengthen their overall security posture. Those findings suggest that organizations can improve their ability to prevent and respond to cyber threats by combining risk assessment with continuous threat monitoring and proactive response strategies. The researchers also highlighted the importance of systems that support ongoing security evaluation, risk tracking, and incident management.

The study emphasizes the need to identify threats, assess vulnerabilities, monitor risks, and prepare for security incidents. These findings are useful for LEXSHIELD because the

platform includes security monitoring and risk management functions to help law firms deal with possible cyber threats.

"A Cybersecurity Risk Management Framework for Organizations"

Saeed, H., Yousafzai, S., & Gill, A. (2021)

Saeed, Yousafzai, and Gill (2021) proposed a cybersecurity risk management framework that helps organizations identify, assess, and manage cybersecurity threats more effectively. The researchers explained that organizations often struggle to understand their security weaknesses because cyber risks involve many factors, including technology, employee behavior, and organizational processes. These challenges make it difficult to develop effective security strategies without a structured approach.

The framework introduced in the study combines important cybersecurity activities such as risk identification, threat analysis, vulnerability assessment, mitigation planning, and continuous security improvement. The authors emphasized that organizations should move away from reacting only after security incidents occur and instead adopt proactive risk management strategies that reduce the chances of future cyberattacks.

This research provides support for using a structured process to identify and manage cybersecurity risks. Its focus on threat analysis, vulnerability assessment, mitigation, and continuous improvement is relevant to LEXSHIELD's security monitoring and risk management features.

"Cybersecurity Incident Response: A Systematic Literature Review"

Ahmad, A., Maynard, S. B., & Shanks, G. (2015)

In their study, Ahmad, Maynard, and Shanks (2015) examined how organizations manage cybersecurity incidents and improve their ability to respond to security threats. Through

a systematic review of existing studies, the researchers analyzed different incident response practices and identified the key factors that contribute to effective cybersecurity incident management. Their review focused on how organizations can strengthen their response capabilities before, during, and after security incidents.

The researchers found that successful incident response depends on having well-defined procedures, effective communication, trained personnel, and continuous improvement based on lessons learned from previous incidents. They also explained that incident response should not only focus on recovering from cyberattacks but should also include investigating the causes of incidents and implementing measures to prevent similar attacks in the future.

The study highlights the importance of having clear procedures, communication, trained users, and lessons learned when dealing with cybersecurity incidents. These findings support LEXSHIELD's incident management functions, which can help law firms respond to security problems and improve their practices after an incident.

"Information Security Risk Management: A Review of the Literature"

Soomro, Z. A., Shah, M. H., & Ahmed, J. (2016)

Based on the review conducted by Soomro, Shah, and Ahmed (2016), information security risk management plays an important role in protecting organizational information from cybersecurity threats. The researchers examined different approaches used by organizations to manage information security risks and found that effective protection requires more than just technical security solutions. It also depends on proper planning, organizational policies, and continuous risk assessment.

The study explains that organizations should regularly identify security vulnerabilities, evaluate possible threats, and apply appropriate security controls to reduce risks. The authors

also highlighted the importance of security awareness among employees, strong governance, and compliance with security standards to ensure that sensitive information remains protected from both internal and external threats.

This study emphasizes that information security risk management requires regular assessment, proper planning, policies, and security controls. Its findings provide a basis for LEXSHIELD's approach to identifying security risks and helping law firms maintain better protection of sensitive information.

"A Review of Cybersecurity Risk Management Frameworks"

Alsmadi, I., Al-Kabi, M., Alsmadi, H., & Al-Kharabsheh, A. (2018)

Alsmadi, Al-Kabi, Alsmadi, and Al-Kharabsheh (2018) reviewed different cybersecurity risk management frameworks that organizations use to identify, evaluate, and reduce cybersecurity risks. The researchers compared several existing frameworks and discussed how structured risk management approaches help organizations strengthen their overall cybersecurity practices. The study also examined how these frameworks support organizations in dealing with changing cyber threats.

The researchers explained that cybersecurity frameworks provide systematic methods for identifying vulnerabilities, assessing risks, implementing security controls, monitoring threats, and continuously improving security practices. They emphasized that because cyber threats constantly evolve, organizations should adopt flexible and adaptive risk management strategies that can respond to new and emerging security challenges. The findings show that effective cybersecurity risk management depends on continuous risk assessment, proactive threat monitoring, and close integration between technical security measures and organizational processes.

The research shows how cybersecurity frameworks can provide organizations with a clear way to identify risks, apply security controls, and monitor threats. This supports LEXSHIELD's use of a structured cybersecurity approach that allows law firms to manage and review security risks more consistently.

"Human Factors in Cybersecurity: A Field-Tested Framework for Designing Resilient, Human-Centered Cybersecurity Systems"

Robinson, Nobles, and Cochran (2025)

The book explains that effective cybersecurity depends not only on technology but also on understanding how people think, behave, and make decisions. The authors point out that many security problems happen because systems are designed without considering human behavior, which often leads to user mistakes, security fatigue, and burnout among cybersecurity professionals. Instead of blaming users for security failures, the book encourages organizations to create systems that are easier to use and that support people in making safe decisions. It presents practical frameworks for reducing human errors, improving incident response, managing mental workload, and designing security controls that match real human behavior. The authors also discuss how human factors engineering and cyber psychology can help organizations predict user behavior, strengthen security operations, and build trust among employees.

Overall, the book emphasizes that combining technical security with human-centered design can create more resilient, reliable, and effective cybersecurity systems that better protect organizations from evolving cyber threats.

The study shows that cybersecurity is also affected by how people use and interact with security systems. This is relevant to LEXSHIELD because its security features should be simple

and understandable, helping lawyers and staff follow secure practices without making the system difficult to use.

"Human Factors in Cybersecurity Awareness"

Sangwan (2024)

The study explains that people play an important role in both causing and preventing cybersecurity threats. Even with continuous improvements in technology, human behavior remains one of the biggest reasons why security incidents happen. The author examines how psychological, behavioral, and social factors influence the way people understand cyber risks and follow security practices. The study also discusses the strengths and weaknesses of cybersecurity awareness programs, including challenges such as keeping people engaged and providing enough resources for training. Using real-world case studies, the paper shows that ignoring the human side of cybersecurity can lead to serious security incidents and data breaches. The author suggests that organizations should combine advanced technology with active human involvement instead of relying on technology alone.

The study concludes that improving cybersecurity awareness requires a balanced approach that includes effective policies, continuous education, good security practices, supportive organizational culture, and cooperation between people, technology, and legal frameworks to better protect against cyber threats.

This study emphasizes that human behavior remains an important part of cybersecurity even when organizations have strong technical controls. Its findings support the need for LEXSHIELD to combine security features with user awareness and clear security practices to reduce mistakes that may lead to security incidents.

F. Cybersecurity Information Sharing, Collaboration, and Knowledge Management

" Cybersecurity Information Sharing Governance Structures: An Ecosystem of Diversity, Trust, and Tradeoffs "

Elaine M. Sedenberg & James X. Dempsey (2018)

According to Sedenberg and Dempsey (2018), cybersecurity information sharing plays an important role in helping organizations improve their ability to detect and respond to cyber threats. The paper explained that successful collaboration depends on trust between organizations, the protection of sensitive information, well-defined policies, and effective information-sharing structures. The authors pointed out that organizations are more willing to share valuable security information when clear guidelines and governance practices are in place. They also emphasized that sharing threat information allows organizations to recognize potential risks more quickly, improve their understanding of emerging cyber threats, and strengthen their incident response capabilities. Overall, the paper suggests that building trust and establishing proper governance are essential for creating effective cybersecurity partnerships and improving organizational resilience against cyberattacks.

The study shows that effective cybersecurity information sharing depends on trust, clear policies, and proper governance. This is relevant to LEXSHIELD because secure sharing of security information and activity records can help law firms improve their awareness of risks while protecting confidential information.

" Knowledge Management for Cybersecurity in Organizations "

Chen, Wang, & Zhang (2021)

The paper explored the relationship between cybersecurity and knowledge management practices in organizations. Chen, Wang, and Zhang (2021) explained that managing and sharing security-related knowledge is an important part of building a strong cybersecurity program. The authors discussed how properly documenting security policies, incident reports, lessons learned,

and best practices allows organizations to respond more effectively to cyber threats. They pointed out that when employees have access to accurate and up-to-date security information, they are better prepared to recognize risks, make informed decisions, and follow appropriate security procedures. The paper also emphasized that effective knowledge management supports continuous learning and helps organizations improve their cybersecurity strategies over time. Overall, the authors concluded that sharing and maintaining cybersecurity knowledge strengthens an organization's ability to respond to threats, improve decision-making, and protect valuable information assets.

This research highlights the value of keeping security policies, incident records, lessons learned, and best practices organized and accessible. These findings support LEXSHIELD's reporting and record-management functions, which can help law firms use previous security information to improve their future practices.

"Cybersecurity Information Sharing: An Overview of Challenges and Solutions"

T. Sommestad, H. Karlzén, & J. Hallberg (2019)

According to Sommestad, Karlzén, and Hallberg (2019), sharing cybersecurity information among organizations plays an important role in improving protection against cyber threats. Their study examined the common challenges and possible solutions related to exchanging information about cyberattacks, security vulnerabilities, and threat patterns. The researchers explained that organizations can strengthen their security by learning from one another, but many are still reluctant to share information because of concerns about confidentiality, privacy, legal responsibilities, and the possibility that sensitive data may be misused.

The study also reviewed different methods used to support cybersecurity information sharing, including technical platforms, information-sharing communities, and governance

mechanisms. The authors emphasized that successful collaboration depends not only on technology but also on clear policies, trusted relationships, and strong organizational commitment to information sharing.

The study explains that organizations can improve cybersecurity by sharing information about attacks, vulnerabilities, and threats, but privacy and confidentiality concerns must also be considered. This supports LEXSHIELD's goal of providing secure ways to manage cybersecurity information while protecting sensitive legal data.

"A Survey of Cybersecurity Information Sharing: Challenges, Solutions, and Future Directions"

Y. Liu, X. Liu, & H. Zhang (2020)

Liu, Liu, and Zhang (2020) conducted a survey that explored the current practices, challenges, and future directions of cybersecurity information sharing. The researchers explained that organizations can improve their ability to defend against cyber threats by exchanging information about malicious activities, security incidents, and system vulnerabilities. However, they also found that many organizations face obstacles that limit effective information sharing, including lack of trust, privacy concerns, inconsistent data formats, and uncertainty about legal responsibilities.

The study reviewed different information-sharing models and technologies, such as cyber threat intelligence platforms, automated information exchange systems, and collaborative security networks. The authors explained that successful information sharing requires standardized procedures, secure communication channels, and cooperation among organizations to ensure that information is exchanged safely and efficiently. The findings show that strengthening cybersecurity information-sharing mechanisms can improve threat detection, support faster incident response, and enhance overall cybersecurity management.

This study shows that sharing information about cyber threats can improve threat detection and incident response. Its discussion of secure communication, standardized procedures, and collaboration is relevant to LEXSHIELD's goal of making security information easier to manage while maintaining proper protection.

"Cyber Threat Intelligence Sharing: A Review of the Current State and Future Research Directions"

A. Tounsi & H. Rais (2018)

In their review, Tounsi and Rais (2018) discussed the importance of cyber threat intelligence sharing in helping organizations strengthen their cybersecurity. The study explained that cyber threat intelligence enables organizations to collect, analyze, and exchange information about cyber threats so they can prepare for attacks and respond more effectively. By sharing threat intelligence, organizations can identify attack patterns, recognize new risks, and make better security decisions.

The researchers examined several cyber threat intelligence frameworks and identified common challenges that affect information sharing. These include poor data quality, lack of trust between organizations, privacy concerns, and difficulties integrating information from multiple sources. The authors emphasized that effective intelligence sharing requires cooperation among organizations, government agencies, cybersecurity providers, and other stakeholders.

The conclusions suggest that improving cyber threat intelligence exchange helps organizations develop proactive cybersecurity strategies, detect threats more quickly, and strengthen their overall defense capabilities.

The research explains how sharing cyber threat information can help organizations recognize attack patterns and respond to new risks. This supports LEXSHIELD's cybersecurity

monitoring approach by showing the value of collecting and using threat information when making security decisions.

"Sharing Cybersecurity Information: A Review of the Challenges and Opportunities"

J. R. Cárdenas, R. González, & M. Fernández (2021)

Cárdenas, González, and Fernández (2021) explained that cybersecurity information sharing has become an important strategy for improving organizational security as cyber threats continue to grow in complexity. The researchers pointed out that it is becoming increasingly difficult for organizations to defend themselves against cyberattacks without collaborating with others. Sharing cybersecurity knowledge, threat intelligence, and security experiences helps organizations improve awareness, identify vulnerabilities, and respond more effectively to security incidents.

The study examined the technical, organizational, and legal factors that influence successful information sharing. The authors found that organizations need to establish trust, protect confidential information, define clear responsibilities, and implement secure information-sharing platforms to support effective collaboration. They also emphasized that good governance practices are necessary to ensure reliable and secure information exchange. The observation show that cybersecurity information sharing is an essential part of modern cybersecurity strategies because it promotes collaboration, improves threat intelligence capabilities, and strengthens organizational resilience.

The research explains how sharing cyber threat information can help organizations recognize attack patterns and respond to new risks. This supports LEXSHIELD's cybersecurity monitoring approach by showing the value of collecting and using threat information when making security decisions.

"Cybersecurity Knowledge Management: A Framework for Improving Organizational Security Awareness"

M. P. Gupta, R. Agrawal, & A. Sharma (2019)

Based on the study by Gupta, Agrawal, and Sharma (2019), effective knowledge management is an important factor in improving cybersecurity awareness and strengthening organizational security. The researchers explained that cybersecurity depends not only on technical security controls but also on the knowledge, awareness, and ability of employees to recognize and respond to security risks. Organizations with well-informed employees are better prepared to prevent cybersecurity incidents caused by human error.

The study proposed that organizations should establish structured knowledge management practices, including documenting cybersecurity policies, sharing lessons learned from previous incidents, maintaining security records, and providing regular employee training. The authors emphasized that continuous knowledge sharing helps build a strong security culture and encourages employees to follow good cybersecurity practices.

The findings indicate that organizations with effective cybersecurity knowledge management are more capable of preventing security incidents, responding to cyber threats, and continuously improving their cybersecurity performance. The researchers concluded that knowledge management should be considered an essential part of cybersecurity governance and organizational resilience.

The study shows that maintaining and sharing cybersecurity knowledge can help employees recognize risks and respond to threats more effectively. This supports LEXSHIELD's goal of organizing security information, records, and reports so that law firms can improve their cybersecurity awareness and practices over time.

2.2 Local Studies

Local studies are research conducted in the Philippines by Filipino researchers or institutions that provide information about issues, practices, policies, and challenges within the country's context. For this study, local research was reviewed to better understand cybersecurity, data privacy, legal compliance, cybercrime laws, and information governance in the Philippines. These studies helped identify existing gaps and provided valuable information that supports the development of the proposed LEXSHIELD: Integrated Cybersecurity and Legal Compliance Platform for Law Firms. To make the presentation more organized and easier to understand, the local studies are grouped into the following categories.

A. Cybersecurity Studies

"Internet Security Awareness of Filipinos: A Survey Paper"

D. Omorog & R. P. Medina (2020)

As internet use continues to grow in the Philippines, understanding how people protect themselves online has become increasingly important. This research examined the level of internet security awareness among Filipinos by reviewing existing studies and survey results related to online behavior and cybersecurity practices. The researchers aimed to determine whether internet users have enough knowledge to recognize common cyber threats and whether they apply safe security practices while using online platforms.

The results showed that many Filipinos are already familiar with basic cybersecurity concepts and understand the importance of protecting their personal information. Most users know that passwords should be kept private and that suspicious websites and messages should be avoided. However, the research also revealed that many internet users still practice unsafe

online habits. Some continue to use weak or repeated passwords, click links without checking their source, ignore software updates, and have difficulty identifying phishing emails and online scams. These behaviors increase the possibility of identity theft, unauthorized access to personal accounts, financial fraud, and other cyber-related crimes.

Another important point discussed in the research is that cybersecurity awareness should continue to improve because cyber threats become more advanced as technology develops. Simply knowing the basic security rules is not enough if users do not consistently apply them in their daily online activities. The researchers emphasized that schools, government agencies, businesses, and internet service providers should work together to promote cybersecurity awareness through seminars, training programs, public campaigns, and educational materials. By increasing public knowledge and encouraging responsible online behavior, Filipinos can become better prepared to protect themselves from different forms of cyberattacks.

The study shows that many Filipinos understand basic cybersecurity practices but still have unsafe online habits. This is relevant to LEXSHIELD because the platform can help users follow safer practices through security controls, reminders, and features that protect sensitive legal information.

"The Current Landscape of Cybersecurity in the Philippines: A Bibliometric Analysis"

Wilber B. Sabado (2025)

Using bibliometric analysis, this research examined how cybersecurity studies in the Philippines have developed over the years. The author analyzed published journal articles, conference papers, and other academic works to determine the growth of cybersecurity research, identify the most common research topics, and examine publication trends. The

purpose of the study was to provide a clear picture of the country's current cybersecurity research landscape and identify areas that still require further investigation.

The analysis showed that cybersecurity has become an increasingly active field of research in the Philippines because more organizations and individuals now rely on digital technologies for communication, business transactions, education, and public services. As cyber threats continue to increase, researchers have also become more interested in studying information security, cybercrime prevention, network security, digital forensics, and cybersecurity education. The growing number of publications indicates that cybersecurity is now recognized as an important field that supports national development and digital transformation.

Despite the increase in research output, the study also found that some important topics have received less attention. These include organizational cybersecurity, cyber defense planning, cloud security, artificial intelligence in cybersecurity, cybersecurity governance, incident response, and risk management. The author explained that these areas are becoming more important because organizations are adopting new technologies that also introduce new security risks.

The research concluded that continued investment in cybersecurity research is necessary to help organizations develop better security practices and strengthen national cybersecurity efforts. It also encouraged stronger collaboration among universities, government agencies, and private companies to produce more research that can address real-world cybersecurity problems and support evidence-based policymaking.

Sabado's study shows that cybersecurity research in the Philippines is growing, while areas such as cybersecurity governance, risk management, and incident response still need

more attention. These findings support the development of LEXSHIELD as a platform that brings cybersecurity monitoring, risk management, and compliance into one system for law firms.

"A Comparative Study of the Philippines in a Global Cybersecurity Context and its Implications on Local Cybersecurity Practices"

David John Buenaventura Dacanay et al. (2024)

To understand the country's position in the global cybersecurity environment, the researchers compared the cybersecurity readiness of the Philippines with that of several other countries. Different factors were examined, including national cybersecurity strategies, government policies, workforce development, digital infrastructure, cybersecurity programs, and the ability to respond to cyber incidents. The comparison was intended to identify the country's strengths while recognizing areas that still need improvement.

The findings showed that the Philippines has made noticeable progress in improving its cybersecurity capabilities through the implementation of national cybersecurity plans, stronger government initiatives, and increased awareness of digital security. Various organizations have also invested in better security technologies and cybersecurity training to protect their digital systems from cyber threats.

Although significant improvements have been made, the study found that many challenges still remain. One major concern is the shortage of qualified cybersecurity professionals who can support organizations in managing cybersecurity risks. The researchers also pointed out that the implementation of cybersecurity policies is not always consistent across different sectors. In addition, many organizations continue to experience difficulties in keeping up with rapidly changing cyber threats because attackers constantly develop more advanced methods of compromising systems and stealing sensitive information.

Based on these observations, the researchers emphasized that cybersecurity should be viewed as a continuous process rather than a one-time effort. Strong cooperation between the government, educational institutions, private companies, and cybersecurity professionals is necessary to improve national preparedness. The study concluded that regular policy improvements, workforce development, investment in cybersecurity technologies, and continuous public awareness programs are important steps toward building a safer digital environment in the Philippines.

This study highlights the need for continuous improvement in cybersecurity because threats and technologies are constantly changing. Its findings support LEXSHIELD's goal of providing law firms with security and compliance features that can be regularly monitored and improved as new risks arise.

"Internet Security and Privacy Awareness Among College Students in the Philippines"

Jim Alves-Foss & Baby Grace Llego (2024)

According to Alves-Foss and Llego (2024), cybersecurity awareness is an essential skill that helps college students protect their personal information and maintain their privacy while using digital devices and the internet. Their study explored whether the age at which students first started using technology, together with their level of technology exposure and frequency of internet use, had an influence on their understanding of cybersecurity and online privacy. To answer this, the researchers surveyed approximately 200 college students from several universities in the Philippines and evaluated both their cybersecurity awareness and their online security practices. After analyzing the collected data, they found that age, technology exposure, and internet usage did not have a significant relationship with the students' level of cybersecurity

awareness. This means that students who had been using technology for many years were not necessarily more knowledgeable about cybersecurity than those with less experience.

The findings suggest that simply spending more time online does not automatically improve a person's understanding of online security. Because of this, the researchers recommended that colleges and universities provide cybersecurity education that is based on the specific learning needs of students. They emphasized that structured training and awareness programs are more effective than relying on technology experience alone to develop safe and responsible online behavior.

The study found that simply having more experience with technology does not necessarily lead to better cybersecurity awareness. This supports the need for LEXSHIELD to provide clear and practical security features that guide users in protecting personal and confidential legal information.

"Cybersecurity Skills in New Graduates: A Philippine Perspective"

John Paul P. Miranda, Marlon I. Tayag, & Joel D. Canlas (2025)

The research conducted by Miranda, Tayag, and Canlas (2025) focused on identifying the cybersecurity knowledge, technical competencies, and workplace skills that employers expect from newly graduated professionals in the Philippines. The researchers reviewed current industry demands and gathered information through surveys involving students, educators, and cybersecurity professionals to determine the skills that are most valuable in today's workforce. The findings revealed that employers highly value graduates who have a solid understanding of cybersecurity concepts and practical technical skills that can be applied to real security problems. However, the study also showed that technical knowledge alone is not enough to succeed in the field. Companies also give equal importance to soft skills such as communication, teamwork, critical thinking, problem-solving, adaptability, and ethical decision-making because

these abilities help employees work effectively with others and respond to different cybersecurity challenges.

The researchers concluded that graduates who possess both technical and interpersonal skills are better prepared for professional responsibilities. Based on these findings, they recommended that Philippine higher education institutions strengthen their cybersecurity curricula by combining technical training with activities that develop professional and personal skills to prepare students for future careers in cybersecurity.

The research emphasizes that cybersecurity work requires both technical knowledge and skills such as communication, problem-solving, and ethical decision-making. This is useful to LEXSHIELD because the platform is intended to support users not only through technical security controls but also through clear processes that make security responsibilities easier to understand and follow.

"CyberSAFE: A Gamified Web Platform with Analytics to Promote Cybersecurity Literacy Among IT Students"

Alfredo L. Calimbo et al. (2025)

In their study, Calimbo and his fellow researchers (2025) developed CyberSAFE, a gamified web-based learning platform designed to improve the cybersecurity literacy of Information Technology students in the Philippines. The platform was created to make cybersecurity education more engaging by combining quizzes, interactive challenges, rewards, progress tracking, and learning analytics into the learning process. These features encouraged students to participate more actively in their lessons while giving instructors useful information about student progress and the topics where additional support was needed. During the implementation of the platform, students became more interested in cybersecurity lessons and

showed a better understanding of online threats, password security, data privacy, phishing attacks, and other safe digital practices.

The researchers also observed noticeable improvements in both the students' cybersecurity awareness and their overall knowledge after completing the learning activities. The findings suggest that students learn more effectively when lessons are interactive and enjoyable. Based on the results, the researchers concluded that combining gamification with learning analytics can improve student engagement, strengthen cybersecurity knowledge, and encourage safer online behavior among Information Technology students.

This study shows that interactive features, progress tracking, and practical activities can improve cybersecurity learning and user engagement. Although LEXSHIELD is designed for law firms rather than students, the findings support the use of clear, interactive, and easy-to-understand security features to encourage users to follow safer practices.

High Awareness, Persistent Vulnerability: Youth Cybersecurity Risk Perception and Behavior"

Arnel R. Rosano et al. (2026)

Rosano et al. (2026) examined the relationship between cybersecurity awareness, risk perception, and protective behavior among Filipino youth in contemporary digital environments. Using a cross-sectional mixed-methods research design, the study collected data from 403 college students to determine whether having knowledge of cybersecurity threats translates into safer online practices. The findings showed that most participants demonstrated a relatively high level of cybersecurity awareness, particularly regarding phishing attacks, online scams, identity theft, and data privacy. However, statistical analyses revealed only weak relationships between cybersecurity awareness and actual protective behaviors, such as verifying suspicious links and practicing safe public Wi-Fi usage. The researchers concluded that awareness alone does not

consistently lead to secure online behavior because many young users continue to make risky decisions despite understanding common cyber threats. The study emphasized that cybersecurity vulnerability among youth is influenced not only by technical knowledge but also by behavioral habits, digital narratives, trust cues, and AI-generated misinformation.

Based on these findings, the researchers recommended that cybersecurity education should move beyond awareness campaigns by incorporating behavior-focused training, experiential learning activities, and realistic cybersecurity scenarios that encourage students to consistently apply safe online practices.

The study found that high cybersecurity awareness does not always result in safe online behavior. This supports LEXSHIELD's use of practical security controls and guidance because users need more than information to consistently protect confidential data and avoid risky actions.

B. Data Privacy and Legal Compliance Studies

"Assessing Compliance of Philippine State Universities to the Data Privacy Act of 2012"

James V. Presbitero & Michelle Renee D. Ching (2018)

This research evaluated how Philippine state universities implement the requirements of the law by examining their institutional policies, privacy procedures, information security measures, and overall privacy management practices. The researchers wanted to determine whether universities have established effective systems for protecting the personal information of students, faculty members, employees, and other stakeholders.

The evaluation revealed that many state universities have already taken important steps to comply with the Data Privacy Act. Several institutions have created privacy policies, assigned

personnel responsible for data protection, implemented security measures, and developed procedures for collecting, storing, and processing personal information. These efforts demonstrate that many universities recognize the importance of safeguarding sensitive information and complying with national privacy regulations.

However, the study also identified several areas that require additional improvement. Some universities continue to experience difficulties in consistently implementing privacy policies, conducting regular employee training, maintaining proper documentation, performing privacy risk assessments, and strengthening information security management. According to the researchers, these challenges may affect an institution's ability to fully comply with the requirements of the law.

The study emphasized that data privacy compliance should be treated as a continuous organizational responsibility instead of a one-time requirement. Universities should regularly review their policies, improve employee awareness, strengthen security controls, and monitor their privacy practices to ensure that personal information remains protected. Continuous improvement will help educational institutions maintain compliance while building greater trust among students, employees, and the public.

The study shows that having privacy policies and security measures is important, but organizations also need regular monitoring, training, and proper documentation to maintain compliance. This is relevant to LEXSHIELD because its compliance features are designed to help law firms monitor their privacy practices and protect confidential client information.

"A Maturity Model Toolkit on Information Governance for Philippine Universities to Aid in Implementing Compliance to the Data Privacy Act of 2012"

Danny Cheng (2021)

This study examined the challenges that Philippine universities face in complying with the Data Privacy Act of 2012 (RA 10173). To address these challenges, the researcher developed an information governance maturity model toolkit specifically designed for higher education institutions. The toolkit provides a structured and step-by-step approach to improving data privacy compliance through practical resources, including guidance documents, self-assessment tools, Business Process Management Notation (BPMN) process flow examples, templates, and implementation advisories. These materials were created to help universities evaluate their current information governance practices and identify areas that require improvement. The toolkit was assessed using an iterative mixed-method approach with the participation of the National Privacy Commission, data protection officers, industry experts, and university stakeholders. The results showed that the toolkit is practical, easy to use, and effective in helping institutions translate regulatory requirements into everyday governance practices.

The researcher concluded that adopting a holistic and maturity-based information governance framework can strengthen data privacy compliance, improve information security, and provide a useful model that can also be adapted by organizations beyond the higher education sector.

The study shows that a structured and step-by-step approach can help organizations assess their current privacy practices and identify areas that need improvement. This supports LEXSHIELD's compliance module, which can help law firms track their compliance status, identify weaknesses, and improve their information governance practices.

"Understanding the Data Privacy Act of 2012: Safeguarding the Filipino's Digital Rights"

Pia Sarah Claveria, Marmelo V. Abante, & Florinda G. Vigonte (2025)

Recognizing that many universities need practical guidance in managing information responsibly, this research introduced an information governance maturity model toolkit to support compliance with the Data Privacy Act of 2012. The researcher designed the toolkit to help educational institutions evaluate the effectiveness of their current information governance practices and identify areas where improvements are needed. Rather than simply checking whether an institution complies with legal requirements, the framework encourages continuous development of governance processes over time.

The maturity model assesses several important areas of information governance, including organizational leadership, policy development, records management, documentation, data handling procedures, privacy compliance, risk management, and information security controls. Through this structured assessment, universities can determine their current level of maturity, recognize weaknesses, and develop practical action plans to improve their governance practices. The toolkit also provides a clear basis for measuring progress as institutions strengthen their information management systems.

Another important contribution of the study is its emphasis on the broader value of information governance. Effective governance not only helps institutions comply with legal requirements but also improves accountability, transparency, operational efficiency, and decision-making. Proper governance ensures that information is collected, stored, shared, retained, and disposed of according to established policies while protecting sensitive data from unauthorized access or misuse.

The research concluded that implementing a structured information governance framework enables universities to continuously strengthen their privacy management practices and improve compliance with the Data Privacy Act of 2012. By regularly assessing and

improving governance processes, institutions can better protect personal information, reduce organizational risks, and build a stronger culture of responsible information management.

The study emphasizes the importance of proper information governance, including policies, documentation, risk management, and security controls, in protecting personal information. These findings support LEXSHIELD's goal of combining data privacy compliance with cybersecurity features to help law firms manage sensitive client data responsibly.

"Perceptions of Students, Faculty and Administrative Staff on the Data Privacy Act: An Exploratory Study"

David Paul R. Ramos (2019)

To better understand how members of educational institutions view data privacy, this exploratory study examined the awareness and understanding of the Data Privacy Act among students, faculty members, and administrative staff. The research aimed to determine how familiar these groups are with their rights and responsibilities under the law and how their level of knowledge may affect the protection of personal information within the institution.

The findings revealed that participants had different levels of awareness regarding the Data Privacy Act. Some respondents were knowledgeable about the law and understood the importance of protecting personal information, while others had only limited knowledge of data privacy principles and legal responsibilities. The study showed that awareness often depended on an individual's role, experience, and exposure to privacy-related activities or training. Because of these differences, some members of the academic community may not fully understand the proper handling, collection, storage, and sharing of personal data.

The research emphasized that protecting personal information requires the participation of everyone in an educational institution, not only those assigned to privacy-related positions.

For this reason, the researcher recommended conducting regular awareness campaigns, seminars, and training programs to improve knowledge of the Data Privacy Act. Strengthening privacy education can help students, faculty, and staff better understand their responsibilities, encourage compliance with privacy regulations, and promote a culture of responsible information management within educational institutions.

The study found that people have different levels of understanding of data privacy and that regular training can improve their knowledge and responsibilities. This is relevant to LEXSHIELD because the system can provide clear privacy procedures and security guidance to help lawyers and staff handle personal and confidential information properly.

***"RA 10173 and its Challenges to Philippine State Universities and Colleges'
Compliance Performance: The Case of Mindanao State University–General Santos City"***
Lumer Jude P. Doce & Michelle Renee D. Ching (2018)

According to Doce and Ching (2018), higher education institutions have an important responsibility to protect the personal information of students, faculty, and staff by complying with Republic Act No. 10173, also known as the Data Privacy Act of 2012. Their study examined how well Mindanao State University–General Santos City complied with the requirements of the law using a qualitative case study approach. The researchers assessed the university's existing data privacy practices and identified the challenges that affected the implementation of its privacy policies. The findings revealed that the university was only partially compliant with the requirements of the Data Privacy Act. The study showed that limited awareness of data privacy policies among employees, insufficient financial and technical resources, and the lack of institutional priority given to privacy programs were the major reasons for the university's low level of compliance. These challenges made it difficult for the institution to fully implement effective privacy measures and protect personal information.

Based on the findings, the researchers emphasized the need for stronger leadership support, continuous awareness campaigns, regular employee training, and better allocation of resources to help universities achieve full compliance with data privacy regulations and improve the protection of sensitive information.

The study identifies limited awareness, lack of resources, and weak institutional support as common challenges in complying with the Data Privacy Act. These findings support LEXSHIELD's purpose of providing a centralized platform that can make privacy compliance easier to monitor and help law firms keep their security responsibilities organized.

"Philippine Data Privacy Law: Is it Implemented in a Private University Library, or Not?"

Rachelle Joy T. Lazo & David Cabonero (2021)

In their study, Lazo and Cabonero (2021) examined how the Philippine Data Privacy Act was being implemented in a private university library and evaluated the level of awareness among the people who regularly use and manage its services. The researchers applied a mixed-methods research design by collecting both quantitative and qualitative data to assess the library's existing privacy practices and determine how well library personnel and users understood data privacy principles. The findings showed that most stakeholders had a satisfactory level of awareness about data privacy and that the library generally complied with the requirements of the Data Privacy Act. However, the researchers also identified several areas that required further improvement, particularly in strengthening privacy procedures, improving documentation, and increasing awareness through regular training programs and information campaigns.

The study concluded that protecting personal information in academic libraries requires continuous education, regular policy reviews, and consistent implementation of data privacy

practices. These efforts can help ensure that libraries continue to provide secure and responsible management of personal data.

The research shows that organizations may have satisfactory privacy practices but still need regular training, documentation, and policy improvements. This supports LEXSHIELD's compliance and record-management features, which can help law firms maintain proper privacy procedures and keep track of their compliance activities.

"Data Privacy Best Practices of a Local Higher Educational Institution: A Model for Governance"

Rushid Jay S. Sancon (2023)

Sancon (2023) explored the data privacy governance practices of Laguna State Polytechnic University to identify effective strategies that support compliance with the Data Privacy Act of 2012. The study focused on examining the university's privacy policies, organizational structure, and management practices to determine how these contributed to protecting personal information and strengthening data privacy governance. The findings revealed that the university had made significant improvements in its privacy program by establishing clear institutional policies, appointing a Data Protection Officer, conducting regular awareness and training activities, and implementing proper privacy management procedures. These initiatives helped the institution improve its ability to protect personal information while meeting the legal requirements of the Data Privacy Act.

The study also showed that having strong leadership support and well-defined governance practices played an important role in maintaining effective data privacy management. Based on the findings, the researcher proposed a governance model that can serve as a useful guide for other higher education institutions in developing, improving, and sustaining their own data privacy programs and compliance efforts.

The study highlights the importance of strong leadership, clear policies, training, and proper data privacy management in maintaining compliance with the Data Privacy Act. These findings are useful to LEXSHIELD because the platform combines governance, compliance monitoring, and cybersecurity practices to help law firms better protect client information.

C. Cybercrime and Philippine Legal Framework Studies

"Safeguarding the Digital Nation: A Narrative Exploration of Roles, Objectives, Cybersecurity Challenges and COMELEAK Case"

Joana Marie Laroco, Florinda G. Vigonte, & Marmelo G. Abante (2025)

Using the COMELEAK incident as the main point of discussion, this paper explored the current state of cybersecurity governance in the Philippines and the challenges faced in protecting government information systems. The research discussed how one of the country's largest data breaches exposed the personal information of millions of registered voters and demonstrated the serious effects of weak cybersecurity practices. Through this case, the authors examined the responsibilities of government agencies in protecting sensitive information and responding to cyber incidents.

The paper also reviewed existing cybersecurity laws, national policies, and government programs that support cybersecurity and digital governance. While these legal measures provide a foundation for protecting information systems, the authors explained that effective implementation remains a major challenge. Factors such as limited technical resources, increasing cyber threats, coordination issues among government agencies, and rapidly changing technology continue to affect the country's ability to respond effectively to cyberattacks.

Another important point discussed in the study is the need for continuous improvement in national cybersecurity efforts. The authors emphasized that preventing future incidents requires

stronger cooperation among government agencies, private organizations, cybersecurity professionals, and the public. They also highlighted the importance of regularly updating cybersecurity policies, strengthening security controls, improving employee awareness, and investing in modern technologies. The study concluded that a proactive and well-coordinated cybersecurity strategy is necessary to better protect government information systems and maintain public trust in digital services.

The study shows that weak cybersecurity practices can lead to serious data breaches and loss of public trust. This is relevant to LEXSHIELD because law firms also handle sensitive information, making strong security controls, incident response, and regular monitoring important for protecting client data.

"Safeguarding Cyberspace: A Comprehensive Analysis of the Cybercrime Prevention Act of 2012 (RA 10175) and Its Role in Philippine Digital Governance"
Antonio Suminig Jr., Marmelo V. Abante, & Florinda G. Vigonte (2025)

A comprehensive review of the Cybercrime Prevention Act of 2012 (Republic Act No. 10175) was conducted to examine its role in addressing cybercrime and supporting digital governance in the Philippines. The study explained the purpose of the law, the different cyber offenses it covers, and the responsibilities of government agencies responsible for enforcing its provisions. It also discussed how the law contributes to creating a safer online environment as more government services, businesses, and individuals depend on digital technologies.

The authors analyzed the implementation of the law by examining its strengths, limitations, and legal issues that have emerged since its enactment. The findings showed that the Cybercrime Prevention Act has become an important legal tool for investigating and prosecuting online crimes such as hacking, identity theft, online fraud, cyber libel, illegal access, and other cyber-related offenses. However, the study also recognized that enforcing cybercrime

laws can be difficult because technology continues to evolve, allowing cybercriminals to develop new methods of attacking digital systems.

In addition, the paper discussed the importance of maintaining a balance between strengthening cybersecurity and protecting the rights of citizens. The authors emphasized that while effective law enforcement is necessary to reduce cybercrime, privacy rights, freedom of expression, and other fundamental rights must also be respected. They concluded that improving digital governance requires continuous updates to cybersecurity laws, stronger cooperation among law enforcement agencies, better public awareness, and responsible use of digital technologies.

The study explains the importance of RA 10175 in addressing cybercrime and protecting digital systems in the Philippines. This supports LEXSHIELD's legal compliance features, particularly its focus on helping law firms manage cybersecurity practices while considering the legal responsibilities related to digital information.

"Cybercrime Prevention Act of 2012 in Practice: Cybersecurity, Controversy, and the Future of Digital Rights in the Philippines"

Alyssa Brucal, Marmelo V. Abante, & Florinda G. Vigonte (2025)

According to Brucal, Abante, and Vigonte (2025), the Cybercrime Prevention Act of 2012 plays an important role in strengthening cybersecurity and addressing the growing number of cybercrimes in the Philippines. Their study reviewed how Republic Act No. 10175 has been implemented and examined its effects on cybersecurity, law enforcement, and the protection of digital rights. The researchers discussed how the law has helped government agencies investigate and prosecute different forms of cybercrime, including cyber libel, identity theft, online fraud, child exploitation, and other internet-related offenses. They also examined several well-known legal cases to demonstrate how the law has been applied in actual situations and how it

has influenced the country's digital legal system. Despite these positive contributions, the study pointed out that some provisions of the law continue to raise concerns about freedom of expression, the right to privacy, and the balance between protecting public safety and preserving individual rights.

The researchers concluded that while RA 10175 remains an important legal framework for combating cybercrime, there is still a need for continuous policy improvements, fair and consistent implementation, and stronger safeguards for digital rights. They emphasized that these measures will help ensure that the law remains effective in addressing future cybersecurity challenges while protecting the rights of internet users in the Philippines.

The study highlights the need to balance cybersecurity, legal enforcement, and the protection of privacy and digital rights. This is relevant to LEXSHIELD because the platform is designed to help law firms strengthen security while also supporting compliance with Philippine cybersecurity and data privacy requirements.

D. Information Governance and Organizational Security

"National Government Agency's Compliance on Data Privacy Act of 2012: A Case Study"

Vicente A. Pitogo (2019)

Focusing on a Philippine national government agency, this case study evaluated how the organization complies with the requirements of the Data Privacy Act of 2012. The assessment examined the agency's privacy policies, security measures, information management practices, and overall compliance with the law. The purpose of the study was to determine whether the agency had established effective systems for protecting the personal information it collects, processes, and stores.

The evaluation showed that the agency had already implemented several privacy-related policies and security measures. However, the overall level of compliance was found to be only partial because of several existing challenges. Among the factors identified were limited employee awareness of data privacy responsibilities, insufficient resources to fully implement privacy programs, and difficulties in consistently applying privacy policies throughout the organization. These issues reduced the agency's ability to achieve full compliance with the requirements of the law.

The researcher explained that compliance with data privacy regulations requires continuous effort and strong organizational support. Employees should receive regular privacy training to improve their understanding of legal responsibilities and proper data handling practices. The study also recommended strengthening privacy policies, improving security controls, allocating sufficient financial and technical resources, and regularly monitoring compliance activities. According to the researcher, these improvements will help government agencies better protect personal information while maintaining public trust and meeting the standards required by the Data Privacy Act of 2012.

The study shows that lack of employee awareness, limited resources, and poor implementation can make data privacy compliance difficult. This is relevant to LEXSHIELD because the platform can help law firms organize their privacy responsibilities, monitor compliance, and protect confidential client information.

"Cross-border Data Regulation for Digital Platforms: Data Privacy and Security"

Atty. Aiken Larisa O. Serzo (2021)

The study discussed how organizations operating in different countries should comply with Philippine data privacy laws while managing personal information that may be stored, processed, or shared outside the country. The research highlighted the growing importance of

international data protection as businesses continue to rely on cloud services, online applications, and global digital platforms.

The paper explained that organizations handling cross-border data must ensure that personal information remains protected throughout its entire lifecycle, regardless of where the data is processed or stored. It emphasized that compliance involves more than simply following legal requirements. Organizations should also establish clear data governance policies, implement strong security measures, manage privacy risks, and adopt responsible information management practices to prevent unauthorized access, data breaches, and misuse of personal information.

Another important point discussed in the study is the need for organizations to remain updated on both Philippine privacy laws and international data protection standards. Since different countries have different legal requirements, organizations must carefully manage cross-border data transfers to avoid legal and security issues. The paper concluded that effective information governance, continuous risk assessment, strong cybersecurity practices, and compliance with applicable privacy regulations are essential for protecting sensitive information and maintaining public confidence in digital services.

The research points out the need to protect personal information wherever it is stored, processed, or shared. This supports LEXSHIELD's goal of helping law firms manage sensitive client data while following privacy requirements and applying proper security measures.

"Implementation of Digital Governance in the Philippine SUCs: Basis for an Enterprise-Level Information System Model"

Allen Paul Esteban, Keno Piad, Isagani Tano, Joseph Espino, Ace C. Lagman, Jayson Victoriano, & Jonilo Mababa (2024)

According to Esteban et al. (2024), digital governance has become an important part of improving the quality of services and decision-making in Philippine State Universities and Colleges (SUCs). Their study examined how digital governance can be strengthened through the development of an enterprise-level information system model that integrates different institutional systems into a single platform. The researchers explained that connecting various information systems can better support instruction, research, extension services, and administrative operations by allowing departments to work together more efficiently.

The findings showed that a unified information system improves communication among offices, supports faster and more accurate data sharing, reduces duplicate records, and enables administrators to make timely and informed decisions. The study also highlighted the importance of having standardized governance policies, system interoperability, and strong information security measures to protect institutional data from unauthorized access and cyber threats. Based on these findings, the researchers concluded that successful digital governance depends on committed leadership, organizational support, compliance with the Data Privacy Act of 2012, and the adoption of enterprise-wide governance frameworks that improve institutional performance and the delivery of public services.

The study found that bringing different information systems together can improve data sharing, reduce duplicate records, and make work more efficient. This relates to LEXSHIELD because the platform also brings cybersecurity, legal compliance, and information management functions into one system for law firms.

"A Maturity Model Toolkit on Information Governance for Philippine Universities to Aid in Implementing Compliance to the Data Privacy Act of 2012 (RA 10173)"

Danny Cheng (2024)

The study conducted by Cheng (2024) introduced an information governance maturity model designed to help Philippine universities improve their compliance with the Data Privacy Act of 2012. The researcher developed a practical toolkit that allows higher education institutions to assess their current governance practices, measure their level of maturity, and identify areas that require improvement. The toolkit covers several important aspects of information governance, including leadership, policy development, records management, privacy compliance, risk management, and information security. It was specifically designed to address the common governance challenges faced by universities while providing a clear and systematic approach for improving their information management practices. To ensure its effectiveness, the maturity model was validated through consultations with data protection officers, regulators, and university representatives, who confirmed its value in supporting institutional governance and privacy compliance.

Based on the findings, the researcher concluded that adopting a maturity-based governance framework can help universities strengthen data security, improve information management, enhance regulatory compliance, and build a more effective and sustainable governance system over time.

The study shows how a structured system can help organizations check their current privacy practices and find areas that need improvement. This is useful to LEXSHIELD because its compliance features are intended to help law firms monitor their compliance status and improve their information governance practices.

"Evaluating Public Information Systems: Enhancing Transparency, Efficiency, and Citizen Engagement in Digital Governance"

Rowena Gracilla, Florinda G. Vigonte, & Marmelo V. Abante (2025)

With the increasing use of digital technologies in government services, evaluating public information systems has become essential for improving digital governance. In this study, Gracilla, Vigonte, and Abante (2025) examined how Public Information Systems contribute to enhancing transparency, efficiency, accountability, and citizen participation in Philippine public institutions. The researchers identified five key areas that should be considered when evaluating these systems: accessibility and inclusiveness, security and data privacy, interoperability, system efficiency, and user experience. These factors were recognized as important in developing reliable, secure, and user-friendly digital government services that meet the needs of both government agencies and the public.

The study also emphasized that effective governance policies should be supported by strong cybersecurity measures and data privacy practices to ensure that public information remains protected while improving the quality of government services. Based on their findings, the researchers concluded that regular system evaluation, continuous modernization, and ongoing improvements in information governance are necessary to strengthen public trust, improve institutional performance, and support the long-term digital transformation of public institutions in the Philippines.

The study highlights security, data privacy, system efficiency, and user experience as important parts of a good information system. These ideas are relevant to LEXSHIELD since the platform is designed to provide law firms with a secure, organized, and easy-to-use system for managing legal information and compliance activities.

E. Cybersecurity Awareness and Compliance Behavior

"Cybersecurity Awareness of College Students in a Private Higher Education Institution"

Buenaventura Alabab, Jonathan Cubol, Michael Angelo Pascual, Ednallyn Ubaldo, Divina Roma Sario, & Marjorie S. Velasco (2023)

According to Alabab et al. (2023), developing cybersecurity awareness among college students is important because they regularly use the internet for learning, communication, and other daily activities. Their study assessed the level of cybersecurity awareness among students in a private higher education institution in the Philippines using a descriptive-correlational research design. A total of 291 students participated in the survey, which measured their knowledge of phishing attacks, identity theft, password security, malware, and other common online threats. The findings showed that most students had a satisfactory level of cybersecurity awareness and understood the importance of protecting their personal information while using digital technologies. However, despite having this knowledge, many students remained vulnerable to phishing attacks, identity theft, and other unsafe online practices. The researchers explained that awareness alone does not always lead to secure online behavior.

Based on these findings, they recommended strengthening institutional cybersecurity awareness programs, integrating cybersecurity topics into academic courses, and conducting regular training activities that encourage students to apply safe online practices in their everyday lives.

This study shows that knowing about cybersecurity threats does not always mean that users will practice safe online behavior. This is relevant to LEXSHIELD because the system can provide security features and reminders that help users apply proper cybersecurity practices when handling sensitive information.

"Internet Security and Privacy Awareness Among College Students in the Philippines"

Jim Alves-Foss & Baby Grace Llego (2024)

In their study, Alves-Foss and Llego (2024) examined the relationship between students' exposure to technology and their level of cybersecurity awareness across several universities in the Philippines. Around 200 college students participated in the survey, which evaluated their understanding of password security, phishing attacks, online privacy, and other important cybersecurity concepts. The findings showed that most students had a good level of cybersecurity awareness and recognized the importance of protecting their personal information while using digital technologies. However, the researchers observed that students' level of awareness differed depending on their technological background and previous cybersecurity training. This suggests that simply having more experience with technology does not always result in stronger cybersecurity knowledge or safer online behavior.

The conclusion suggests the importance of continuously integrating cybersecurity education into higher education curricula through practical lessons, awareness campaigns, and hands-on activities to further improve students' digital safety and responsible online practices.

The study highlights the importance of continuous cybersecurity education and practical training in improving users' online safety. This supports LEXSHIELD's goal of helping users become more aware of cybersecurity and privacy while using a system that handles confidential legal information.

***"Cybersecurity Awareness, Data Privacy Practices, and Behavioral Compliance
Among College Students in a Provincial University in the Philippines"***

Enrique B. Picardal Jr. (2026)

The study conducted by Picardal (2026) explored the relationship between cybersecurity awareness, data privacy practices, and behavioral compliance among 200 college students in a provincial university in the Philippines. Using a quantitative descriptive-correlational research

design, the researcher examined whether students who were knowledgeable about cybersecurity also practiced safe and responsible online behavior. The findings revealed that most students demonstrated a high level of cybersecurity awareness and had a good understanding of common online threats, cybersecurity risks, and the importance of protecting personal information. However, their actual data privacy practices and behavioral compliance were only at a moderate level, showing that many students did not consistently apply what they knew when using the internet in their daily activities.

The study also identified a significant but weak positive relationship between cybersecurity awareness and behavioral compliance, suggesting that awareness alone is not enough to guarantee secure online behavior. Based on these findings, the researcher recommended behavior-focused cybersecurity education, stronger institutional data privacy policies, and regular practical training to help students develop safer, more responsible, and consistent online habits.

Picardal's findings show that cybersecurity awareness alone may not be enough to ensure proper data privacy practices. This supports LEXSHIELD's use of security controls and compliance features that can help users consistently follow safe practices when managing personal and confidential information.

"Cybersecurity Awareness and Cybersecurity Behavior of High School Students in Davao City: A Mediation Role of Perceived Behavioral Control"

Kenneth Budiongan & Ramil Carballo (2024)

Using the Theory of Planned Behavior as its framework, Budiongan and Carballo (2024) examined the relationship between cybersecurity awareness and the online behavior of high school students in Davao City. The researchers also investigated whether perceived behavioral

control, or the students' confidence in their ability to practice safe online behavior, influenced this relationship. The findings showed a significant positive relationship between cybersecurity awareness and secure online practices, indicating that students with higher awareness were more likely to behave responsibly while using digital technologies. In addition, students who believed they had greater control over their online actions were more confident in applying safe cybersecurity practices in their everyday internet use.

These results suggest that knowledge becomes more effective when students also have the confidence to apply what they have learned. Based on the findings, the researchers concluded that increasing awareness alone is not enough to promote safe online behavior. They recommended integrating cybersecurity education into secondary school curricula, strengthening digital citizenship programs, and providing practical learning activities that encourage students to practice responsible, safe, and ethical online behavior.

This study suggests that users are more likely to practice safe cybersecurity behaviors when they have both knowledge and confidence in applying it. The findings are useful to LEXSHIELD because the platform should not only provide security measures but also make these measures clear and easy for users to follow when handling legal information.

2.3 Related Literature

Related literature includes books, journals, research papers, articles, government publications, and other reliable sources that are related to the topic of the study. These sources provide useful information, ideas, and findings from previous researchers that help explain the research topic. It allows researchers to gather information from different sources and use it to support their research. Reading related literature helps researchers understand important

concepts, compare the ideas of different authors, and see how previous studies are connected to their own research.

To make the discussion organized and easy to understand, the related literature in this study is divided into different topics. Each topic covers an important area that is connected to the research and presents literature from different sources that supports the study.

A. Cybersecurity

Cybersecurity has become an important part of every organization because many business activities now depend on digital systems and online services. As organizations continue to store confidential information electronically, they also become more vulnerable to cyber threats such as malware, phishing attacks, ransomware, unauthorized access, and data breaches. Because of these risks, organizations need to implement security measures that protect their systems and information from cybercriminals.

According to the National Institute of Standards and Technology (2024), cybersecurity is not limited to preventing attacks. It also involves identifying security risks, protecting information assets, detecting security incidents, responding to cyberattacks, and recovering from them. Following these practices helps organizations reduce security risks, maintain business operations, and protect confidential information. For law firms, cybersecurity is especially important because they manage legal documents, client records, contracts, and other sensitive information that must remain secure at all times.

The NIST guidelines are relevant to LEXSHIELD because they provide a clear approach to identifying, protecting, detecting, responding to, and recovering from cybersecurity risks. These practices support the security features of LEXSHIELD, especially in monitoring threats and protecting confidential legal information.

According to the Cybersecurity and Infrastructure Security Agency (2023), cybersecurity is essential for protecting critical systems, sensitive information, and digital services against constantly evolving cyber threats. Organizations are encouraged to implement layered security measures such as multi-factor authentication, strong password policies, regular software updates, employee cybersecurity awareness training, and continuous monitoring. These security practices reduce the likelihood of successful cyberattacks and improve organizational resilience.

CISA's recommendations support LEXSHIELD's use of basic but important security practices such as strong passwords, multi-factor authentication, security awareness, and continuous monitoring. These measures can help law firms reduce common security risks and better protect client information.

IBM (2024) reported that cyberattacks continue to increase in frequency and financial impact worldwide. Data breaches often result in operational disruption, financial losses, legal liabilities, and reputational damage. The report also found that organizations implementing strong security controls, artificial intelligence-assisted security monitoring, and incident response planning significantly reduce both the cost and recovery time associated with security incidents. These findings highlight the importance of investing in proactive cybersecurity measures.

Their study is relevant to LEXSHIELD because it shows that cybersecurity involves more than just technical protection. This supports the platform's approach of combining security controls, user activities, organizational processes, and legal compliance in one system.

Von Solms and Van Niekerk (2013) argued that cybersecurity extends beyond protecting computer systems and includes safeguarding people, organizational processes, information, and

technology from cyber-related risks. They emphasized that effective cybersecurity requires cooperation between management, employees, and technical personnel. Organizations should establish comprehensive security strategies that address both technological vulnerabilities and human behavior to improve overall cyber resilience.

Their study is relevant to LEXSHIELD because it shows that cybersecurity involves more than just technical protection. This supports the platform's approach of combining security controls, user activities, organizational processes, and legal compliance in one system.

Craigen, Diakun-Thibault, and Purse (2014) defined cybersecurity as the collection of technologies, policies, processes, guidelines, and practices designed to protect cyberspace, users, organizations, and information systems from cyber threats. The authors emphasized that cybersecurity aims to preserve the confidentiality, integrity, and availability of information while ensuring that organizations can continue operating securely despite cyber risks. They further explained that cybersecurity is a multidisciplinary field involving technical, legal, organizational, and human factors that work together to protect digital environments. These concepts support the need for integrated cybersecurity solutions capable of protecting sensitive legal information handled by law firms.

The study supports the idea that cybersecurity involves technology, policies, processes, legal requirements, and human factors. This is closely related to LEXSHIELD since the platform combines cybersecurity features with legal compliance tools to provide more complete protection for law firms.

B. Information Security

Information security focuses on protecting information from unauthorized access, disclosure, alteration, or destruction. It applies not only to electronic records but also to printed documents and other forms of information used by an organization. Data privacy is concerned with protecting personal information and ensuring that it is collected, processed, stored, and shared responsibly. Organizations that collect personal information are expected to protect it against unauthorized access, disclosure, or misuse.

As mentioned by the National Privacy Commission (2016), organizations should implement organizational, physical, and technical security measures to safeguard personal data throughout its lifecycle. Good privacy practices help organizations comply with legal requirements while building trust among clients and stakeholders. Since law firms regularly process sensitive client information, maintaining strong data privacy practices is necessary to protect client rights and preserve confidentiality.

The NPC guidelines are relevant to LEXSHIELD because they emphasize the use of organizational, physical, and technical safeguards when handling personal information. These practices support the platform's goal of helping law firms protect client data and follow the requirements of the Data Privacy Act.

The National Institute of Standards and Technology (2020) explained that information security requires organizations to implement a comprehensive set of security and privacy controls to protect information systems and organizational operations. These controls include access control, identification and authentication, audit logging, incident response, risk assessment, and system monitoring. NIST emphasized that implementing these safeguards helps organizations preserve the confidentiality, integrity, and availability of information while

reducing cybersecurity risks. For law firms, applying these controls strengthens the protection of confidential client records and supports compliance with legal and regulatory requirements.

The NIST guidelines provide security controls that can support features of LEXSHIELD, such as access control, user authentication, audit logs, incident response, and security monitoring. These controls can help law firms protect confidential records and manage cybersecurity risks more effectively.

The International Organization for Standardization (2022) stated that organizations should establish appropriate information security controls to protect information assets from unauthorized access, disclosure, modification, and destruction. ISO/IEC 27002 provides guidance on implementing controls related to access management, cryptography, physical security, asset management, logging and monitoring, incident management, and information protection. Applying these controls helps organizations strengthen their information security management systems and improve resilience against information security risks. These recommendations are particularly valuable for law firms that regularly process confidential legal information.

ISO/IEC 27002 supports LEXSHIELD by providing practical security controls for protecting information from unauthorized access, changes, loss, or disclosure. Its recommendations can guide the security features of the platform, particularly in access management, logging, monitoring, incident handling, and protection of legal information.

C. Data Privacy

Data privacy refers to the protection of personal information and the right of individuals to control how their personal data is collected, used, processed, stored, and shared by organizations. It emphasizes that personal information should be handled in a lawful, fair, and

transparent manner and only for legitimate purposes. Organizations are also expected to collect only the data that is necessary and to implement appropriate safeguards to prevent unauthorized access, disclosure, alteration, or misuse of personal information.

According to the Organization for Economic Co-operation and Development (OECD) (2013), organizations should follow internationally recognized privacy principles, including collection limitation, purpose specification, use limitation, security safeguards, openness, individual participation, and accountability. These principles provide guidance for responsible data processing and help organizations strengthen the protection of personal information. For law firms, maintaining strong data privacy practices is essential because they regularly collect and process highly sensitive personal information and confidential legal records. Protecting this information helps preserve client trust, uphold professional ethics, and support compliance with applicable data protection regulations.

The OECD privacy principles are relevant to LEXSHIELD because they provide basic guidelines for collecting, using, and protecting personal information responsibly. These principles support the platform's compliance features by helping law firms manage client data properly, protect privacy, and maintain accountability.

The European Union (2016) explained that personal data must be processed lawfully, fairly, and transparently while ensuring that it is collected only for specified, explicit, and legitimate purposes. The General Data Protection Regulation (GDPR) also requires organizations to implement appropriate technical and organizational measures to protect personal data against unauthorized access, accidental loss, destruction, or disclosure. In addition, organizations are expected to demonstrate accountability by complying with data protection principles throughout the data processing lifecycle. These principles are applicable to

law firms because they regularly process confidential client information and must ensure that personal data is handled responsibly.

The GDPR principles support LEXSHIELD's goal of promoting proper handling of personal data throughout its processing. Its focus on lawful processing, transparency, security, and accountability can help guide the platform's privacy and compliance features for protecting sensitive client information.

The National Privacy Commission (2022) emphasized that organizations processing personal information must adopt privacy management programs and implement appropriate organizational, physical, and technical security measures to protect personal data. The Commission explained that protecting personal information is essential to safeguarding individuals' privacy rights while promoting trust in digital services. Organizations are likewise encouraged to ensure that employees understand their responsibilities regarding data privacy and information protection.

The NPC emphasizes the need for privacy management programs, security measures, and employee awareness when handling personal information. This is relevant to LEXSHIELD because the platform can help law firms monitor privacy practices, organize compliance activities, and encourage responsible handling of client data.

The Organization for Economic Co-operation and Development (2013) explained that organizations should establish privacy policies based on internationally recognized principles such as collection limitation, purpose specification, use limitation, security safeguards, openness, individual participation, and accountability. The framework encourages organizations to implement responsible data governance practices that protect personal information while supporting the free flow of information across jurisdictions. Applying these principles helps

organizations improve transparency, strengthen public trust, and ensure responsible data processing. For law firms, adherence to these principles contributes to the protection of sensitive client records and compliance with privacy regulations.

The OECD framework supports LEXSHIELD by emphasizing responsible data governance, transparency, security safeguards, and accountability. These principles can guide the platform in helping law firms manage sensitive client records while maintaining proper privacy practices and compliance with applicable regulations.

D. Data Privacy Act of 2012 (Republic Act No. 10173)

The Data Privacy Act of 2012 serves as the primary law governing the protection of personal information in the Philippines. Under Republic Act No. 10173 (2012), organizations are required to collect, process, store, and share personal information in a lawful and responsible manner while protecting the privacy rights of individuals. The law also requires organizations to implement appropriate organizational, technical, and physical security measures to prevent unauthorized processing, accidental disclosure, or misuse of personal information.

Compliance with the law helps organizations avoid legal penalties and strengthens public confidence in how personal information is managed. For law firms, following the Data Privacy Act is important because they routinely process confidential client records and sensitive legal documents.

The National Privacy Commission (2022) explained that the Data Privacy Act of 2012 protects the fundamental human right to privacy while ensuring the free flow of information to promote innovation and economic growth. The Commission emphasized that organizations processing personal information must implement appropriate organizational, physical, and

technical security measures to safeguard personal data against unauthorized access, accidental disclosure, alteration, destruction, and other unlawful processing. Compliance with these requirements enables organizations to strengthen data protection and maintain public trust. For law firms, these measures are essential because they regularly handle confidential client records and sensitive legal information.

The NPC's guidelines are relevant to LEXSHIELD because they stress the need for proper organizational, physical, and technical measures to protect personal information. This supports the platform's goal of helping law firms protect confidential client data and monitor their privacy and security practices.

Republic Act No. 10173 (2012) provides the legal framework governing the protection of personal information in the Philippines. The law requires organizations to process personal information lawfully, implement appropriate safeguards, respect the rights of data subjects, and ensure accountability in handling personal data. It also establishes the responsibilities of personal information controllers and processors, as well as penalties for violations involving unauthorized processing, negligent access, improper disposal, and unlawful disclosure of personal information. Compliance with these legal obligations is particularly important for law firms because they process sensitive personal and legal information that requires a high level of protection.

Republic Act No. 10173 provides the legal basis for LEXSHIELD's data privacy and compliance features. Its requirements on lawful data processing, security safeguards, data subject rights, and accountability guide the platform in helping law firms handle personal and confidential legal information properly.

E. Cybercrime Prevention Act of 2012 (Republic Act No. 10175)

The Cybercrime Prevention Act of 2012 provides the legal framework for addressing crimes committed through computers, networks, and other digital technologies. As mentioned by Republic Act No. 10175 (2012), the law covers offenses such as illegal access, computer-related fraud, identity theft, data interference, cyber libel, and other forms of cybercrime. It also supports the investigation and prosecution of cybercriminals while promoting cybersecurity in both the public and private sectors. Organizations are encouraged to strengthen their cybersecurity measures to reduce the risk of cybercrime and protect their digital resources. Law firms can benefit from understanding this law because it supports the protection of confidential legal information and digital evidence.

The Cybercrime Prevention Act is relevant to LEXSHIELD because it provides the legal basis for addressing cybercrimes such as illegal access, fraud, identity theft, and data interference. It supports the platform's goal of helping law firms protect their digital resources, confidential legal information, and electronic records from cyber-related threats.

The Council of Europe (2001) explained that the Budapest Convention establishes an international legal framework for combating cybercrime by harmonizing criminal laws, improving investigative procedures, and strengthening international cooperation among countries. The Convention addresses offenses involving illegal access, system interference, data interference, computer-related fraud, child exploitation, and copyright violations. It also encourages countries to develop effective legal mechanisms for investigating cybercrimes while respecting human rights and the rule of law. These principles are relevant to law firms because cybercrime investigations often involve digital evidence and cross-border electronic communications that require secure handling and legal compliance.

The Budapest Convention highlights the importance of proper laws, investigation procedures, and cooperation in dealing with cybercrime. This supports LEXSHIELD's focus on

secure handling of legal information and digital evidence, especially when law firms deal with cybersecurity incidents and electronic records.

According to INTERPOL (2023), cybercriminals continue to exploit phishing, ransomware, business email compromise, and online fraud to target organizations worldwide. INTERPOL emphasized that effective cybercrime prevention requires continuous monitoring, timely reporting of incidents, secure information sharing, and collaboration between law enforcement agencies and private organizations. Strengthening organizational cybersecurity helps reduce opportunities for cybercriminals and improves incident response. These recommendations are valuable for law firms because they frequently manage confidential legal documents and sensitive client information that may be targeted by cybercriminals.

INTERPOL's findings show that threats such as phishing, ransomware, and online fraud continue to affect organizations. This is relevant to LEXSHIELD because its cybersecurity features can help law firms monitor security risks, respond to incidents, and protect sensitive client and case information.

F. Information Security Governance

Information security governance helps organizations ensure that cybersecurity supports business goals and legal responsibilities. It involves leadership, policy development, risk management, accountability, and continuous improvement. As written in International Organization for Standardization (2020), information security governance enables organizations to manage security risks while ensuring that security activities remain aligned with organizational objectives. Strong governance also promotes better decision-making, improves compliance, and encourages management involvement in protecting organizational information. For law firms, effective governance helps ensure that security policies are consistently implemented and that confidential client information receives appropriate protection.

The study is relevant to LEXSHIELD because it shows the importance of having clear policies, risk management, accountability, and management support in protecting information. These ideas support the platform's goal of helping law firms organize their security responsibilities and consistently protect confidential client information.

G. ISO/IEC 27001 Information Security Management System

ISO/IEC 27001 is one of the most widely recognized international standards for information security management. As stated by the International Organization for Standardization (2022), the standard provides organizations with a structured approach for identifying risks, implementing security controls, monitoring security performance, and continuously improving information security. Following ISO/IEC 27001 allows organizations to establish an Information Security Management System (ISMS) that helps protect sensitive information and demonstrate compliance with recognized security standards. Organizations that adopt this framework are generally better prepared to manage cyber risks and improve their overall cybersecurity posture.

ISO/IEC 27001 supports LEXSHIELD by providing a structured way to identify security risks, apply controls, monitor security, and improve protection over time. Its risk-based approach can help guide the platform's cybersecurity and compliance features for law firms.

According to PECB (2023), ISO/IEC 27001 enables organizations to establish a comprehensive Information Security Management System that protects the confidentiality, integrity, and availability of information. The framework promotes risk-based decision-making, continuous improvement, leadership commitment, and regular performance evaluation to ensure that security controls remain effective. Organizations implementing ISO/IEC 27001 are better positioned to manage cybersecurity risks, meet regulatory requirements, and strengthen

organizational resilience. These principles are especially applicable to law firms because they manage highly sensitive legal and personal information.

The study highlights the importance of protecting the confidentiality, integrity, and availability of information through proper risk management and continuous improvement. This is useful to LEXSHIELD because law firms handle sensitive legal and personal information that needs strong and consistent protection.

H. Compliance Management

Compliance management refers to the process of ensuring that organizations follow applicable laws, regulations, policies, and industry standards. It includes identifying legal obligations, monitoring compliance activities, documenting evidence, and correcting compliance issues when necessary. As mentioned by the International Organization for Standardization (2021), an effective compliance management system promotes accountability, improves organizational governance, and reduces legal and operational risks. For organizations that manage sensitive information, such as law firms, compliance management helps maintain legal responsibilities while supporting secure and ethical business operations.

The study shows that compliance management helps organizations identify their legal responsibilities, monitor compliance, and address problems when they occur. This directly relates to LEXSHIELD's compliance features, which are intended to help law firms keep track of their legal and security requirements.

The International Federation of Accountants (2022) stated that effective compliance management enables organizations to identify regulatory obligations, implement internal controls, monitor compliance activities, and address non-compliance promptly. A strong compliance program improves organizational transparency, accountability, and stakeholder

confidence while reducing legal and operational risks. Integrating compliance into organizational governance also supports ethical decision-making and long-term sustainability. For law firms, these practices help ensure that legal services are delivered in accordance with applicable laws and professional standards.

The study emphasizes the importance of internal controls, compliance monitoring, and timely action when organizations fail to meet requirements. These ideas support LEXSHIELD's purpose of helping law firms manage compliance activities, maintain proper records, and reduce legal and operational risks.

I. Role-Based Access Control (RBAC)

Role-Based Access Control (RBAC) is a security approach that limits access to information based on the responsibilities of individual users within an organization. Instead of giving every user the same level of access, permissions are assigned according to specific job roles. This helps reduce unauthorized access, protects confidential information, and simplifies user management. Ferraiolo et al. (2016) explained that RBAC improves information security by ensuring that users can access only the information needed to perform their assigned duties. In law firms, implementing RBAC helps protect confidential legal documents by restricting access to authorized lawyers and staff members.

This study is relevant to LEXSHIELD because it shows how user access can be controlled based on their assigned roles. This supports the platform's use of different permissions for lawyers, staff, administrators, and clients to protect confidential legal information.

According to the InterNational Committee for Information Technology Standards (INCITS) (2012), the RBAC standard provides a formal model for assigning system permissions based on organizational roles rather than individual users. The standard defines concepts such as role hierarchies, separation of duties, and permission assignment to ensure that users can access

only the information necessary for their responsibilities. Implementing standardized RBAC improves security, simplifies user administration, and reduces the risk of unauthorized access. For law firms, applying RBAC helps protect confidential legal documents by ensuring that access is limited to authorized lawyers and staff.

The RBAC standard provides a basis for organizing user permissions in LEXSHIELD. Its concepts can help the system ensure that users only access the legal records and functions needed for their responsibilities.

Sandhu et al. (1996) introduced the Role-Based Access Control model as an efficient method for managing user permissions within organizations. The authors explained that RBAC simplifies access management by assigning permissions to roles instead of individual users, making security administration more efficient and scalable. The model also supports the principle of least privilege by ensuring that users receive only the access required to perform their duties. These principles are highly applicable to law firms, where lawyers, paralegals, and administrative staff require different levels of access to confidential client information.

This study supports LEXSHIELD's access control feature by showing how assigning permissions to roles can make security management easier and more organized. This is useful for law firms since different employees need different levels of access to client and case information.

According to the Open Worldwide Application Security Project (OWASP) (2021), access control is one of the most critical aspects of application security because failures in authorization can allow unauthorized users to access or modify sensitive information. OWASP recommends implementing role-based authorization, enforcing the principle of least privilege, validating permissions on every request, and regularly reviewing user roles. Organizations that follow these

practices reduce the likelihood of unauthorized access and privilege escalation attacks. For law firms, these recommendations strengthen the protection of confidential legal records and client information stored within legal management systems.

The study is useful to LEXSHIELD because it highlights the risks of weak access control and excessive user privileges. Its recommendations support the use of role-based access and least-privilege permissions to prevent unauthorized users from viewing or changing confidential legal records.

According to Microsoft (2023), Role-Based Access Control enables organizations to manage permissions by assigning users to predefined roles with specific privileges. This approach simplifies security administration, improves accountability, and minimizes excessive user permissions that could lead to unauthorized access. Microsoft also emphasized that RBAC supports the principle of least privilege and enhances organizational security by ensuring users have access only to resources necessary for their work. For law firms, RBAC helps secure legal case files, client records, and confidential documents against unauthorized access.

This source is relevant to LEXSHIELD because RBAC provides a simple way to manage what each user can access in the system. Applying this approach can help law firms control access to case files, client records, and other sensitive documents while making user management easier.

J. Audit Trail

An audit trail is a chronological record of activities performed within an information system. It records user actions such as logins, file access, data modifications, and administrative activities. Maintaining an audit trail helps organizations monitor system usage, investigate security incidents, identify unauthorized activities, and demonstrate compliance with legal and

regulatory requirements. As found by Kent and Souppaya (2006), audit logs are an important component of information security because they support accountability, incident investigation, and continuous monitoring. For law firms, audit trails provide evidence of system activities and help protect confidential client information by ensuring transparency and accountability.

This source is relevant to LEXSHIELD because audit trails allow the system to keep a record of user activities such as logins, file access, and changes to information. This can help law firms monitor system use, investigate suspicious activities, and maintain accountability when handling confidential client records.

According to the Center for Internet Security (2021), organizations should collect, retain, and regularly review audit logs to detect unauthorized activities and support timely incident response. Proper log management enables security teams to identify unusual behavior, investigate cybersecurity incidents, and demonstrate compliance with security policies. Protecting audit logs from unauthorized modification further strengthens the integrity of security investigations. For law firms, effective audit trail management helps protect sensitive legal records and ensures accountability for system activities.

The study supports LEXSHIELD's audit trail feature by emphasizing the importance of collecting, protecting, and reviewing system logs. Applying these practices can help the platform detect unusual activities and provide useful records when investigating security incidents.

The European Union Agency for Cybersecurity (2023), emphasized that organizations should maintain comprehensive logging and monitoring mechanisms to improve cybersecurity resilience. Security logs provide valuable information for detecting cyber threats, analyzing incidents, and supporting digital forensic investigations. ENISA also recommended implementing centralized log management and continuous monitoring to strengthen organizational security. These practices are especially important for law firms because they help protect confidential

legal information and provide evidence during security investigations.

This source is useful to LEXSHIELD because logging and monitoring can help identify possible threats and provide evidence during security investigations. These functions can strengthen the platform's ability to monitor activities involving sensitive legal information.

K. Legal Case Management System

A Legal Case Management System is a technology-based solution designed to help law firms and legal organizations organize, manage, and monitor case-related information efficiently. It provides a centralized platform for storing legal documents, tracking case progress, managing schedules, recording client information, and improving communication among legal professionals. According to Brooks (2020), legal technology solutions such as case management systems support law firms by improving document organization, automating routine tasks, and enhancing the management of legal information. Furthermore, integrating security features such as access control, audit trails, and data protection mechanisms is essential because legal case management systems store confidential client information and sensitive legal documents.

For law firms, implementing a secure Legal Case Management System helps improve operational efficiency while ensuring that client records are protected. The integration of cybersecurity, data privacy, and compliance practices within the system supports responsible information management and helps organizations meet legal and ethical obligations in handling confidential legal information.

This study is relevant to LEXSHIELD because it shows how legal case management systems can help organize documents, client information, and case activities. LEXSHIELD applies this concept while adding cybersecurity and compliance features to provide better protection for sensitive legal information.

According to the American Bar Association (2023), law firms should adopt legal technology that enhances the secure management of client information, legal documents, case records, and communications. The ABA emphasized that legal professionals have an ethical responsibility to understand the benefits and risks associated with technology and to implement reasonable safeguards that protect client confidentiality. Secure legal case management systems improve efficiency while helping law firms comply with their professional and ethical obligations.

The ABA's discussion supports LEXSHIELD because it emphasizes the responsibility of law firms to use technology while protecting client confidentiality. This relates to the platform's goal of combining case management with security measures that help protect legal records and client information.

According to Clio (2024), cloud-based legal case management systems have become essential tools for improving law firm efficiency, document organization, client communication, and secure information sharing. The report highlighted that firms adopting integrated legal management platforms experience better workflow efficiency and improved client service while maintaining secure access to legal records through authentication and permission-based controls. These capabilities are especially valuable for law firms handling sensitive client information.

This source is relevant to LEXSHIELD because integrated case management systems can make it easier for law firms to organize documents, manage cases, and communicate with clients. These ideas support LEXSHIELD's case management functions, while its added cybersecurity and compliance features provide another layer of protection for confidential information.

2.4 Synthesis of the Reviewed Literature and Studies

Based on the foreign studies, local studies, and literature reviewed, it is clear that cybersecurity is becoming more important as organizations continue to use digital technologies in their daily operations. Many researchers explained that cyber threats such as hacking, phishing, ransomware, malware, and data breaches can cause serious problems, especially for organizations that handle confidential information. Since law firms manage sensitive client records, legal documents, and personal information, they need stronger security measures to protect their data.

The reviewed studies also showed that using cybersecurity frameworks, security policies, and risk management practices can help organizations reduce security risks. Many authors recommended following standards such as the NIST Cybersecurity Framework and ISO/IEC 27001 because they provide clear guidelines for protecting information and responding to cyber incidents. Other studies also highlighted the importance of following privacy and data protection laws to avoid legal issues and protect clients' information.

The local studies supported these findings by showing that many organizations in the Philippines still experience cybersecurity challenges. Some organizations have limited security resources, while others lack proper security awareness and compliance with existing regulations. These studies suggested that organizations should improve their security practices by using technology, providing employee training, and following government regulations such as the Data Privacy Act of 2012.

Although many studies discussed cybersecurity, legal compliance, and information security, most of them focused on these topics separately. Only a few studies presented a system that combines cybersecurity and legal compliance into one platform, especially for law

firms in the Philippines. This shows that there is still a need for a system that can help law firms manage both security and compliance in one place.

Because of this gap, the researchers developed **LEXSHIELD: Integrated Cybersecurity and Legal Compliance Platform for Law Firms**. The system is designed to help law firms protect confidential information, monitor cybersecurity risks, manage legal compliance, and improve the security of their daily operations. By combining these features into one platform, LEXSHIELD aims to provide a more organized, secure, and efficient way of managing legal data while helping law firms comply with Philippine laws and cybersecurity standards.

CHAPTER 3

METHODOLOGY

3.1 Research Design

This research design focuses on creating, improving, and evaluating the LEXSHIELD system. It helps ensure that the platform meets the needs of lawyers, clients, and administrators while providing secure and reliable legal services.

This study employed a Developmental Research Design because it focuses on the design, development, implementation, and evaluation of the proposed system, **LEXSHIELD: Integrated Cybersecurity and Legal Compliance Platform for Law Firms**. Developmental research is appropriate because the study aims to create a functional web-based platform that addresses cybersecurity risks while improving legal service management and regulatory compliance within law firms.

The study began by identifying the requirements of lawyers, clients, and administrators through interviews, surveys, document analysis, and a review of existing legal management systems. The collected information served as the basis for designing and developing the system. After development, the system underwent comprehensive testing and evaluation to determine whether it satisfies the functional, security, usability, and compliance requirements of its intended users.

3.2 System Development Methodology

Agile Software Development Methodology

The Agile methodology allowed the researchers to develop LEXSHIELD in small stages. It also helps improve the system by making changes based on testing results and user feedback throughout the development process.

The Agile methodology consists of the following phases:

1. Requirements Analysis

- In this phase, the researchers identified the needs and requirements of the proposed system. Information were gathered from the target users through surveys, interviews, and related studies. The researchers analyzed the collected data to determine the features and functions that should be included in LEXSHIELD. One of the identified requirements is an email verification feature for newly registered clients to validate their accounts before they can access the system. The system requirements served as a guide for the design and development process. This phase helps make sure that the system meets the needs of its intended users.

2. System Design

- In this phase, the researchers created the overall design of the proposed system. The system structure, user interface, database, and workflow were planned based on the gathered requirements. The researchers also prepared diagrams and layouts to show how the system will work. The researchers also designed an email verification process using SMTP through PHPMailer to verify newly created client accounts. This phase serves as a guide for the development of LEXSHIELD. A well-planned design helps make the system easier to develop and use.

3. Software Development

- In this phase, the researchers developed the LEXSHIELD system based on the approved system design and identified requirements. The planned features and modules were implemented using the selected programming languages, database, and development tools. Security features such as user authentication, CSRF protection, email verification, role-based access control, malware scanning, phishing URL scanning, hashing, session guard, rate limiter and input validation were also integrated into the system. The completed modules were combined and prepared for testing to ensure that the system functioned as intended. This phase transformed the system design into a functional web-based application.

4. Testing

- In this phase, the researchers tested the developed system to make sure that all features work properly. Different testing procedures were conducted to check the system's functionality, security, and performance. The researchers tested whether only verified email accounts are allowed to access the system. Any errors or problems found during testing will be fixed by the researchers. The system also checked to ensure that it meets the requirements identified during the earlier phases. This phase helps ensure that LEXSHIELD is reliable and ready for evaluation.

5. Deployment (Prototype)

- In this phase, the completed prototype of LEXSHIELD was prepared for presentation and evaluation. The researchers installed and demonstrated the system to the selected users so they could test its features and functions. Feedback and suggestions were collected to identify areas that need

improvement. Any necessary revisions were made to enhance the prototype. This phase helps determine if the system is ready for its intended purpose.

3.3 System Architecture

This section explained how the different parts of LEXSHIELD work together to process data and provide services to users. The system used a three-tier client-server architecture to separate the user interface, system functions, and database. This structure helps improve the system's security, performance, and maintenance. It also allows each layer to perform its specific role without affecting the others.

The three main layers of the system are:

1. Presentation Layer

- This is the part of the system that users interact with. It includes the web pages, forms, and dashboards where users can access the features of LEXSHIELD and perform different tasks.

2. Application Layer

- This layer handles the main functions of the system. It processes user requests, applies the system's logic, and manages features such as cybersecurity monitoring, compliance checking, verification and user authentication.

3. Database Layer

- This layer stores and manages all the data used by the system. It keeps user accounts, legal documents, compliance records, system logs, and other important information organized and secure.
-

3.4 Tools and Technologies Used

This section explained the software and technologies that were used to develop the LEXSHIELD system. These tools helped the researchers design, build, test, and manage the system throughout the development process. Each technology has a specific purpose in creating a functional and secure platform.

- **Frontend: HTML, CSS, JavaScript**

These technologies were used to create the user interface of LEXSHIELD. They helped design the web pages, forms, dashboards, and other visual elements that users will interact with. They were also able to make the system responsive and easy to use.

- **Backend: PHP / JavaScript API / Node.js**

The backend technology handled the main functions of the system. It processes user requests, manages system features, connects to the database, and ensures that the system works properly. It will also support the security mechanisms and overall performance of LEXSHIELD.

- **Database: MySQL**

MySQL was used to store and manage the data of the system. It kept records such as user accounts, legal documents, compliance information, and other important information. This helps organize and protect the data stored in LEXSHIELD.

4. Development Tools: Visual Studio Code, Git, GitHub, XAMPP, Composer, phpMyAdmin, Google Chrome, Microsoft Edge, PHPMailer (SMTP)

These development tools were used throughout the creation of the LEXSHIELD system. They helped the researchers write, test, debug, and manage the source code efficiently. They also supported database management, version control, and team collaboration during development. This helps improve account security and prevents fake or invalid registrations. Using these tools improved the quality and organization of the project. Together, they helped ensure that the system is developed successfully and functions as expected.

3.5 Data Gathering

This section explained the methods that were used to collect the information needed to develop LEXSHIELD. The collected data helped the researchers understand the needs of the target users, identify cybersecurity concerns, and determine the features that should be included in the system. The information gathered also served as the basis for the design and development of the proposed platform.

The data gathered were used in the following ways:

- **Interviews**

Interviews were conducted with lawyers, legal professionals, and IT experts to gather information about their experiences and needs. The researchers also ask questions about their current workflow, cybersecurity practices, and challenges they encounter. The information collected helped improve the design and functionality of LEXSHIELD.

- **Surveys and Questionnaires**

Surveys and questionnaires were distributed to the target users to collect their opinions and suggestions. The researchers gathered information about the features they need, the problems they commonly experience, and their expectations for the proposed system. The responses were used to support the development of LEXSHIELD.

- **Document Analysis**

Document analysis was conducted by reviewing related laws, cybersecurity standards, research studies, and other references. The researchers used these documents to understand legal requirements and cybersecurity practices that should be applied in the system. This helped ensure that LEXSHIELD follows relevant regulations and best practices.

- **Review of Existing Systems**

The researchers reviewed existing legal management and cybersecurity systems to understand how they work. Useful features, strengths, and weaknesses were identified and analyzed. The information gathered improved the design of LEXSHIELD and avoided common problems found in similar systems.

3.6 Testing Procedures

This section explained the testing methods that were used to evaluate the quality of LEXSHIELD. The researchers conducted different tests to make sure that the system works properly, performs its intended functions, and provides a secure and reliable platform for users. The results of these will helped identify and fix any problems before the system is evaluated.

- **Unit Testing**

Unit testing was conducted to check whether each feature or module of LEXSHIELD works correctly on its own. The researchers tested each part of the system separately to identify and fix any errors before combining them with other modules.

- **System Testing**

System testing was conducted to evaluate the complete LEXSHIELD platform. The researchers checked whether all features and functions work as expected and whether the system meets the identified requirements.

- **Security Testing**

Security testing was performed to examine the security features of LEXSHIELD. The researchers verified that user accounts, documents, and other sensitive information are properly protected against common security threats and unauthorized access.

-

- **User Acceptance Testing (UAT)**

User Acceptance Testing (UAT) was conducted by allowing the intended users to use the system and evaluate its features. The researchers collected their feedback on the system's

usability, functionality, and overall performance. The suggestions gathered during this phase were used to improve the final version of LEXSHIELD.

3.7 Evaluation Criteria

This evaluation determined whether LEXSHIELD meets its intended objectives and provides a functional, secure, and user-friendly platform for its intended users. The system was evaluated based on the following criteria:

- **Functionality**

This criterion evaluated whether the features and functions of LEXSHIELD operate correctly and meet the system requirements. It ensured that users can perform their intended tasks successfully.

- **Usability**

This criterion measured how easy the system is to learn and use. It evaluated whether users can navigate the interface and complete tasks efficiently with minimal difficulty.

- **Security**

This criterion assessed how well LEXSHIELD protects user accounts, legal documents, and other sensitive information. It evaluated security features such as authentication, access control, and data protection.

- **Compliance Accuracy**

This criterion evaluated whether LEXSHIELD supports the legal compliance and cybersecurity requirements identified in the study. It was verified that the system follows applicable Philippine laws, standards, and project requirements.

- **Efficiency**

This criterion measured how quickly and effectively LEXSHIELD performs its functions. It evaluated the system's response time, processing speed, and overall performance while handling user requests.

SUPPORTING DIAGRAMS (Included within Chapter 3)

Data Flow Diagram (DFD – Level 0)

This diagram shows how data moves between the users, the LEXSHIELD system, and the database. It provided a simple overview of how information is entered, processed, stored, and returned by the system. It also helps guide the development of LEXSHIELD.

User → LEXSHIELD System → Database

- **Public Visitor** → Views the landing page, browses the lawyer directory, views lawyer profiles and ratings, registers an account, receives an email verification link, verifies the account, and logs in.
- **Client** → Logs in, manages appointments, uploads case documents, sends messages with attachments, submits payment proofs, views payment history, manages case files, and updates profile information.

- **Lawyer** → Logs in, manages professional profile, handles assigned cases, reviews client documents, manages appointments, and communicates securely with clients and administrators.
- **Administrator** → Manages lawyers and clients, reviews payments, monitors audit logs and security activities, configures system settings, sends announcements, and oversees overall system operations.
- **LEXSHIELD System** → Authenticates users, sends account verification emails using SMTP through PHPMailer, processes requests, manages cases, appointments, documents, messages, payments, notifications, audit logs, role-based access, and security features such as CSRF protection and input validation and session guard.
- **Database (MySQL)** → Stores user accounts, lawyer and client profiles, case records, appointments, legal documents, messages, payment records, notifications, audit logs, security logs, and system settings.

Data Flow

- The system sends a verification email to newly registered clients. The account will only be activated after the user successfully verifies the email address.
- Users submit login credentials and other system requests.
- The system authenticates users based on their assigned roles (Administrator, Lawyer, or Client).
- Before processing any user request, the system performs security validation to ensure that only authorized and legitimate requests are accepted.
- The system processes appointments, case information, documents, messages, payments, and profile updates.
- The system stores and retrieves information from the MySQL database.

- The system returns the requested information, notifications, or confirmations to users.
 - All user activities and security events are recorded through audit logging to support monitoring and reporting.
-

Entity Relationship Diagram (ERD)

The **Entity Relationship Diagram (ERD)** shows the database structure of LEXSHIELD. It illustrated the main entities, their attributes, and how they are related to each other. This helped the researchers organize and manage the system's data efficiently.

Entities:

- **User** (UserID, Name, Email, Password, Role, StatusVerificationToken, EmailVerified)
- **LawyerProfile** (LawyerID, UserID, BarNumber, Specialization, Biography, Experience)
- **ClientProfile** (ClientID, UserID)
- **Case** (CaseID, ClientID, LawyerID, Status, Description)
- **Appointment** (AppointmentID, ClientID, LawyerID, Date, Status)
- **Message** (MessageID, SenderID, ReceiverID, Attachment)
- **Payment** (PaymentID, ClientID, Amount, Proof, Status)
- **Document** (DocumentID, CaseID, FileName, FilePath)
- **AuditLog** (LogID, UserID, Activity, Timestamp)
- **Notification** (NotificationID, UserID, Message, Status)

Relationships:

- **User** has one **LawyerProfile** or one **ClientProfile**.
 - **ClientProfile** creates **Case** records.
 - **LawyerProfile** manages **Case** records.
 - **ClientProfile** books **Appointments** with **LawyerProfile**.
 - **User** sends and receives **Messages**.
 - **ClientProfile** submits **Payments**.
 - **Case** contains **Documents**.
 - **User** generates **AuditLogs**.
 - **User** receives **Notifications**.
 - **Administrator** manages **Users**, **LawyerProfiles**, **Payment**
-

Use Case Diagram

The **Use Case Diagram** shows the different users of LEXSHIELD and the actions they can perform within the system. It helped the researchers understand how users interact with the system and its main features.

Actors:

- **Public Visitor**
- **Client**
- **Lawyer**

- **Administrator**

Use Cases:

Public Visitor

- View Landing Page
- Browse Lawyer Directory
- View Lawyer Profiles
- Register Account
- Verify Email
- Login

Client

- Verify Email (First Login Only)
- Login Account
- Manage Profile
- Browse Lawyers
- Book Appointments
- Manage Case Files
- Upload Case Documents
- Send and Receive Messages
- Submit Payment Proof
- View Payment History
- Receive Notifications

Lawyer

- Login Account
- Manage Professional Profile
- View Assigned Cases
- Manage Case Files
- Review Client Documents
- Manage Appointments
- Send and Receive Messages
- Receive Notifications

Administrator

- Manage Users
- Manage Lawyers
- Manage Clients
- Review and Approve Payments
- View Payment Records
- Monitor Audit Logs
- Monitor Security Activities
- Configure System Settings
- Send Notifications
- Generate Reports

CHAPTER 4

SYSTEM DEVELOPMENT, IMPLEMENTATION, TESTING, EVALUATION, RESULTS, AND DISCUSSION

4.2 Overview of the Developed System

The **LEXSHIELD: Integrated Cybersecurity and Legal Compliance Platform for Law Firms** is a web-based legal management system developed to provide secure and efficient legal services while helping law firms protect sensitive information and comply with cybersecurity and data privacy requirements. The system serves as a centralized platform where administrators, lawyers, clients, and public visitors can perform their respective tasks through a secure and user-friendly interface.

Unlike traditional legal management systems that mainly focus on case handling and document management, LEXSHIELD integrates cybersecurity features with legal service management. The system includes secure user authentication, email verification using PHPMailer through SMTP, role-based access control, audit logging, document management, secure messaging, payment management, appointment scheduling, and compliance support. These features help protect confidential legal information while improving the efficiency of law firm operations.

The system was developed to assist law firms in managing client records, legal documents, appointments, communications, and compliance-related activities within a single platform. By combining legal management functions with cybersecurity measures, LEXSHIELD

improves data security, reduces administrative workload, and supports compliance with applicable Philippine laws and cybersecurity standards.

The major objectives of the developed system are as follows:

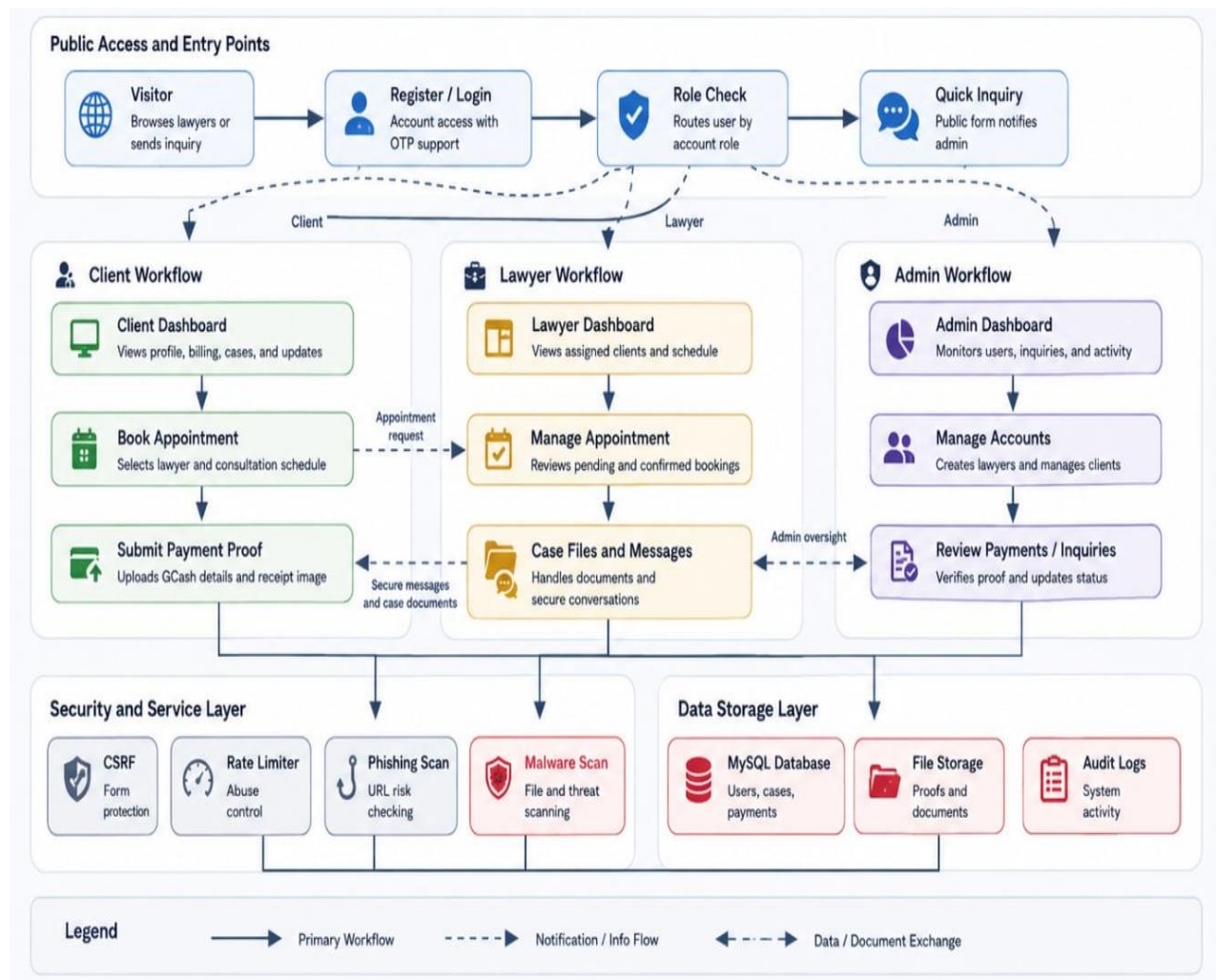
- Provide a secure and integrated platform that addresses the cybersecurity and legal compliance management needs of law firms.
- Offer a simple, user-friendly, and responsive interface that enables lawyers, clients, and administrators to access and manage the system efficiently.
- Integrate modules for cybersecurity monitoring, data privacy management, and compliance tracking based on applicable laws and regulations.
- Monitor system activities, detect potential security risks, and provide updates related to cybersecurity and compliance status.
- Provide an administrative dashboard that enables authorized users to manage system operations, generate reports, review security information, and support decision-making.
- Maintain a secure and centralized database for storing and managing client records, legal documents, case information, and other essential system data while ensuring data protection.
- Deliver a system that meets the required standards for security, usability, performance, reliability, and effectiveness in supporting cybersecurity and legal compliance management.

The development of LEXSHIELD followed modern web application development practices by implementing responsive user interface design, secure database management using MySQL, server-side processing with PHP, version control through Git and GitHub, and development tools such as Visual Studio Code, XAMPP, phpMyAdmin, Composer, and PHPMailer. The system also incorporates security mechanisms such as authentication, email verification, CSRF

protection, data encryption, phishing scanner, malware scanner, hashing, input validation, and audit logging to enhance system reliability and data protection.

The completed system consists of several interconnected modules that work together to provide secure legal services. These modules include user registration and authentication, lawyer profile management, client management, case management, appointment scheduling, document management, secure messaging, payment management, notification services, audit logging, and administrative functions.

Figure 4.1 illustrates the overall workflow of the developed LEXSHIELD system.



The workflow begins when a public visitor accesses the website and registers for an account. After registration, the system sends an email verification link through PHPMailer using SMTP. Once the email address has been verified, the user can log in and access the features assigned to their role. Clients may browse lawyer profiles, book appointments, upload case documents, exchange secure messages, submit payment proofs, and monitor their case information. Lawyers can manage their professional profiles, review assigned cases, communicate with clients, manage appointments, and update case records. Administrators oversee the entire system by managing users, monitoring audit logs, reviewing payment transactions, configuring system settings, and maintaining overall system security. All transactions and activities are securely stored in the MySQL database and are processed through the application layer before being presented to the users through the web interface.

4.3 System Development Methodology

The development of **LEXSHIELD: Integrated Cybersecurity and Legal Compliance Platform for Law Firms** adopted the **Agile Software Development Methodology**, an iterative software development approach that emphasizes continuous planning, development, testing, and improvement. Agile was selected because it allowed the researchers to develop the system in small, manageable phases while incorporating testing and user feedback throughout the development process. This approach ensured that the developed system met the functional, security, and usability requirements identified during the study.

The Agile methodology enabled the researchers to develop LEXSHIELD incrementally. Each phase focused on completing specific system modules, verifying their functionality, and making the necessary improvements before proceeding to the next phase. This process

minimized development errors, improved software quality, and ensured that each component functioned correctly before system integration.

The development process consisted of the following phases:

Requirements Gathering

The first phase involved gathering information from lawyers, legal professionals, and prospective users through interviews, surveys, document analysis, and a review of related legal management and cybersecurity systems. The researchers identified the functional and non-functional requirements of LEXSHIELD, including user authentication, email verification, role-based access control, secure document management, appointment scheduling, messaging, payment management, audit logging, and compliance monitoring. The collected information served as the foundation for designing and developing the system.

System Design

After identifying the system requirements, the researchers designed the overall architecture of LEXSHIELD. The database structure, user interface, process flow, and system modules were planned to ensure efficient operation and data security. The researchers also prepared the Data Flow Diagram (DFD), Entity Relationship Diagram (ERD), and Use Case Diagram to serve as guides during software development. In addition, the email verification process using PHPMailer through SMTP and the role-based access mechanism were incorporated into the system design.

Software Development

During this phase, the researchers implemented the planned system using PHP, HTML, CSS, JavaScript, MySQL, XAMPP, Composer, PHPMailer, Git, GitHub, and Visual Studio Code. The system modules were developed incrementally, including user registration and authentication, lawyer and client profile management, appointment scheduling, case

management, document management, secure messaging, payment management, notification services, audit logging, and the administrative dashboard. Security features such as user authentication, CSRF protection, session guard, rate limiter, input validation, data encryption, hashing, phishing and malware scanning, and role-based access control were also implemented to enhance system security.

Testing

Each completed module underwent unit testing before being integrated into the entire system. Integration testing ensured that the interaction between modules functioned correctly, while system testing verified the overall performance of the application. User Acceptance Testing (UAT) was later conducted with selected respondents to validate whether the application met user expectations.

Deployment

Following successful testing, the completed prototype of LEXSHIELD was prepared for presentation and evaluation. The system was configured using a local server environment through XAMPP and connected to the MySQL database. The researchers demonstrated the developed platform to the selected evaluators, who tested its features and provided feedback. Minor revisions and improvements were implemented before the final version of the system was completed.

Maintenance

The researchers considered future maintenance activities to ensure the system's continued reliability. These activities include correcting software defects, updating security mechanisms, improving system performance, enhancing existing features, and implementing additional functionalities based on user feedback and future cybersecurity requirements.

Figure 4.2 presents the Agile Software Development Life Cycle adopted in the study.

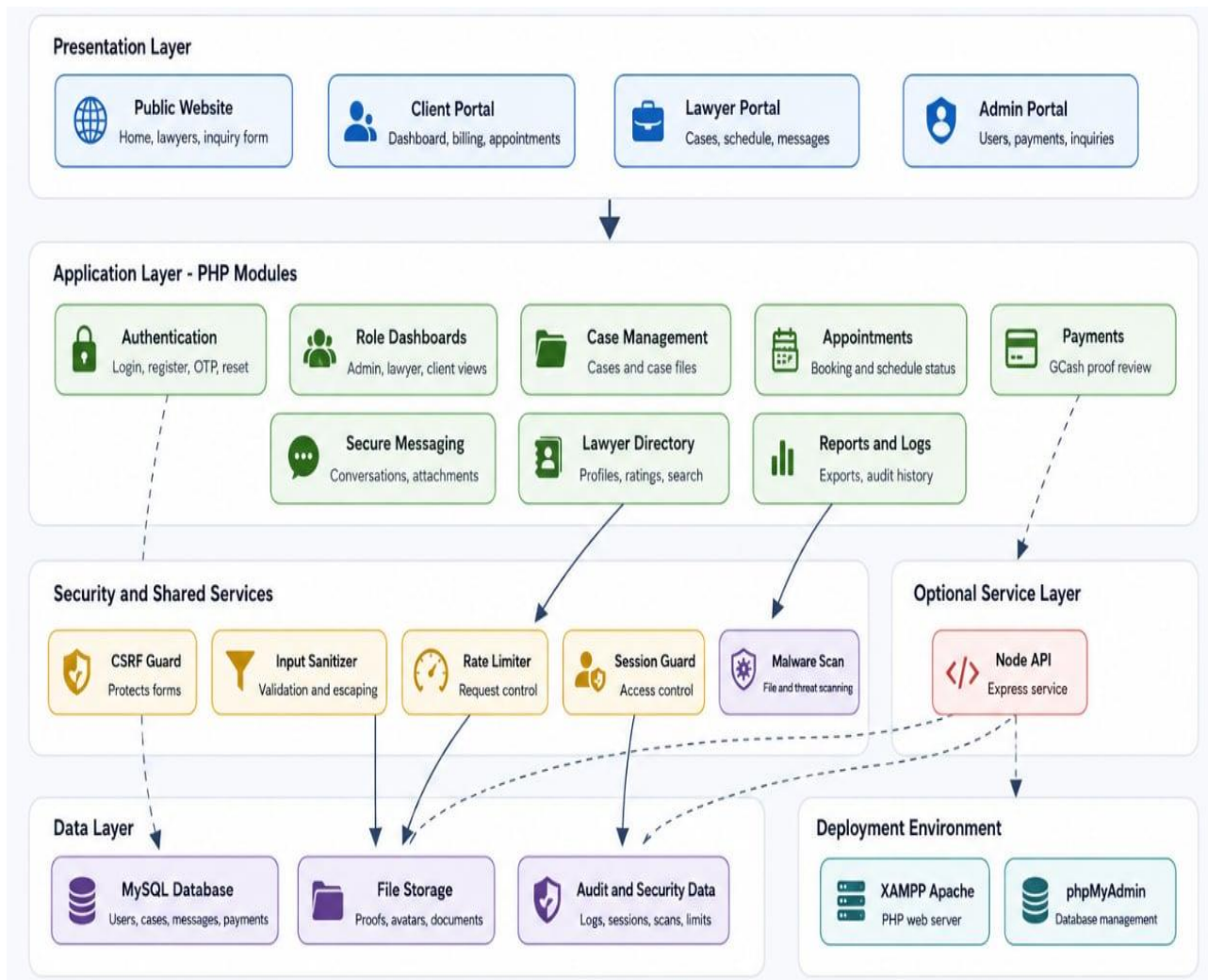


4.4 System Architecture

The **LEXSHIELD: Integrated Cybersecurity and Legal Compliance Platform for Law Firms** utilizes a **Three-Tier Client-Server Architecture**, which consists of the Presentation Layer, Application Layer, and Data Layer. This architecture separates the user

interface, system processing, and database management to improve the system's security, maintainability, and performance.

Figure 4.3 illustrates the architecture of the developed system.



The architecture consists of the following components:

Presentation Layer

The presentation layer serves as the interface between the users and the LEXSHIELD system. It allows users to access the system's features through a web browser using desktop computers, laptops, tablets, or smartphones.

Major interface components include:

- Landing Page
- Login and Registration
- Client Dashboard
- Lawyer Dashboard
- Administrator Dashboard
- Appointment Management
- Case Management
- Document Management
- Messaging System
- Payment Management
- Notifications

Application Layer

The application layer contains the business logic that processes user requests, validates input, manages system operations, and enforces security features and mechanisms. It acts as the communication bridge between the presentation layer and the database.

Its primary responsibilities include:

- User Authentication
- Email Verification
- Role-Based Access Control
- CSRF Protection
- Input Validation
- Session guard

- Case Management
- Appointment Management
- Secure Messaging
- Payment Processing
- Audit Logging
- Notification Management
- Compliance and Security Monitoring

Data Layer

The data layer stores and manages all information required by the system using a centralized MySQL database. It maintains data integrity while ensuring secure storage and retrieval of information.

Stored information includes:

- User Accounts
- Lawyer Profiles
- Client Profiles
- Case Records
- Appointments
- Legal Documents
- Messages
- Payment Records
- Notifications
- Audit Logs

The three-tier architecture enables secure communication between the application and database while protecting confidential legal information through authentication, role-based access control, and other implemented security mechanisms.

4.5 Hardware and Software Requirements

The successful implementation of **LEXSHIELD: Integrated Cybersecurity and Legal Compliance Platform for Law Firms** requires appropriate hardware and software resources. These specifications ensure that the system performs efficiently when processing user requests, managing legal records, storing documents, and implementing cybersecurity features.

4.5.1 Hardware Requirements

Table 4.1 presents the minimum and recommended hardware specifications for running the LEXSHIELD system.

Table 4.1 Hardware Requirements

Component	Minimum Specification	Recommended Specification
Processor	Intel Core i3 or equivalent	Intel Core i5 or higher
Memory (RAM)	4 GB	8 GB or higher
Storage	256 GB SSD	512 GB SSD or higher
Monitor	1366 × 768 Resolution	Full HD (1920 × 1080)

Component	Minimum Specification	Recommended Specification
Internet Connection	10 Mbps	50 Mbps or higher
Mobile Device	Android 10 or later	Android 13 or later

The recommended hardware configuration provides sufficient processing capability to support user authentication, document management, appointment scheduling, secure messaging, and other system functions while maintaining reliable performance during daily operations.

4.5.2 Software Requirements

Table 4.2 presents the software requirements used in the development and operation of the LEXSHIELD system.

Table 4.2 Software Requirements

Software	Purpose
Windows 10/11	Operating System
Visual Studio Code	Source Code Editor
PHP, Python, Node.js	Backend Development
HTML, CSS, JavaScript	Frontend Development
MySQL	Database Management System
XAMPP	Local Development Server

Software	Purpose
phpMyAdmin	Database Administration
Composer	PHP Dependency Management
PHPMailer	Email Verification through SMTP
Git	Version Control
GitHub	Source Code Repository and Collaboration
Google Chrome / Microsoft Edge Web Browser	

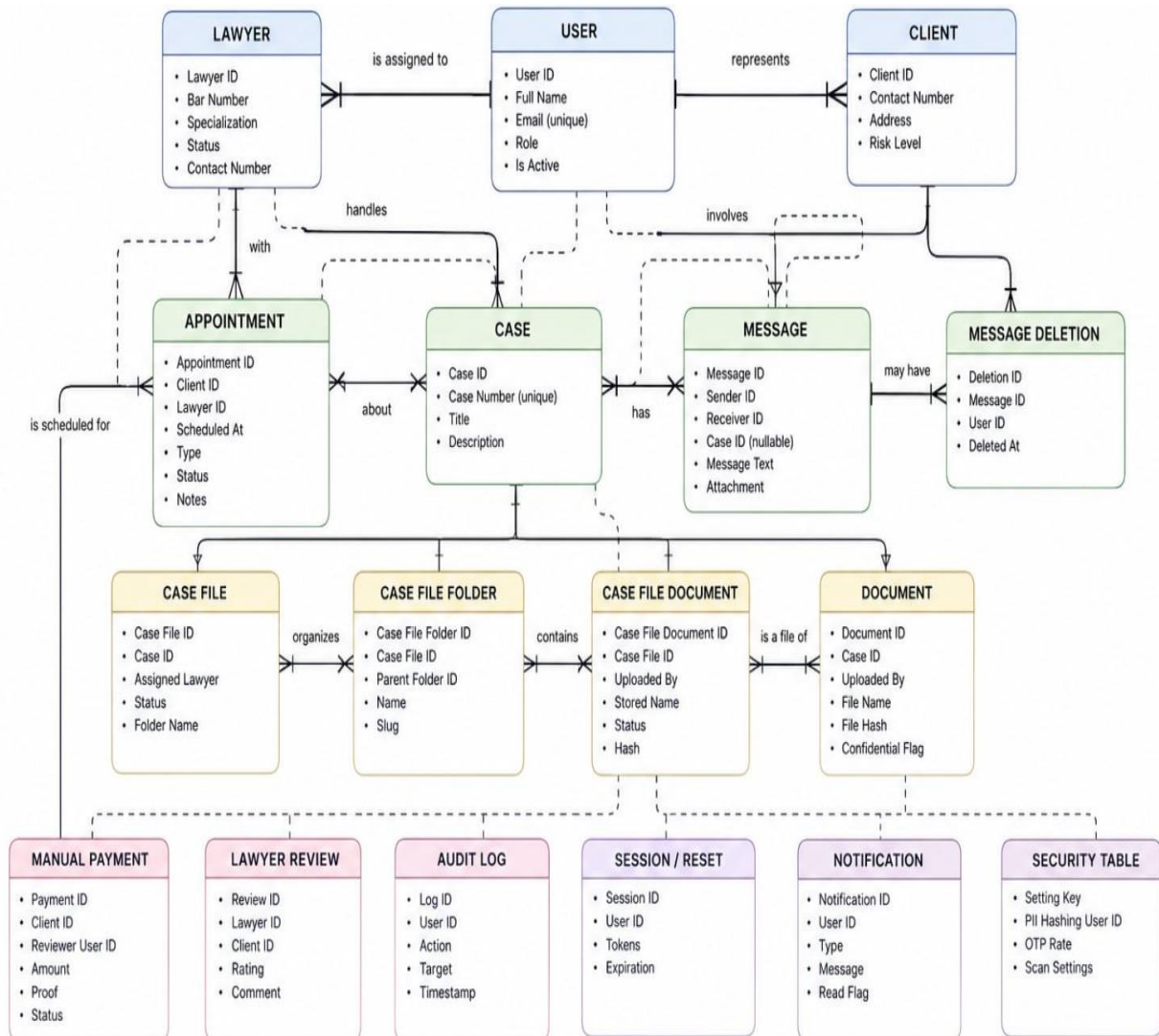
The selected software tools were chosen because of their compatibility, reliability, and suitability for developing a secure web-based legal management system. These technologies supported the implementation of user authentication, email verification, database management, cybersecurity features, and the overall functionality of the system.

4.6 Database Design

The database serves as the foundation of the LEXSHIELD system by organizing and securely storing user accounts, legal records, case information, appointments, documents, messages, payment records, notifications, and audit logs. A well-designed database ensures efficient data retrieval, minimizes data redundancy, and maintains the integrity and security of sensitive legal information.

The researchers implemented a relational database model using MySQL. Database normalization techniques were applied to establish relationships among tables, reduce duplicate data, and improve data consistency and overall system performance.

Figure 4.4 presents the Entity Relationship Diagram (ERD) of the developed system.



The primary entities of the database include:

- User

- LawyerProfile
- ClientProfile
- Case
- Appointment
- Document
- Message
- Payment
- Notification
- AuditLog

These entities work together to support the core functions of LEXSHIELD, including user management, case management, appointment scheduling, document handling, secure communication, payment processing, notification services, and activity monitoring. The relationships among these entities enable the system to maintain data integrity while providing efficient access to information required by lawyers, clients, and administrators.

Users Table

This table stores the account credentials and basic information of all system users, including administrators, lawyers, and clients.

Field	Description
user_id	Primary Key
fullname	Complete Name
email	User Email Address

Field	Description
password	Encrypted Password
role	User Access Level
status	Account Status
verification_token	Email Verification Token
email_verified	Email Verification Status

Lawyer Profiles Table

This table stores the professional information of registered lawyers.

Field	Description
lawyer_id	Primary Key
user_id	Foreign Key
bar_number	Lawyer's Bar Number
specialization	Area of Legal Practice
biography	Professional Background
experience	Years of Experience

Client Profiles Table

This table stores additional information about registered clients.

Field	Description
-------	-------------

client_id	Primary Key
-----------	-------------

user_id	Foreign Key
---------	-------------

Cases Table

This table stores legal case information managed by lawyers and clients.

Field	Description
-------	-------------

case_id	Primary Key
---------	-------------

client_id	Foreign Key
-----------	-------------

lawyer_id	Foreign Key
-----------	-------------

status	Case Status
--------	-------------

description	Case Description
-------------	------------------

Appointments Table

This table stores appointment schedules between lawyers and clients.

Field	Description
-------	-------------

appointment_id	Primary Key
----------------	-------------

Field	Description
client_id	Foreign Key
lawyer_id	Foreign Key
appointment_date	Scheduled Date and Time
status	Appointment Status

Messages Table

This table stores communication records between lawyers, clients, and administrators.

Field	Description
message_id	Primary Key
sender_id	Foreign Key
receiver_id	Foreign Key
attachment	Attached File
message	Message Content
sent_at	Date and Time Sent

Payments Table

This table stores payment information submitted by clients.

Field	Description
payment_id	Primary Key
client_id	Foreign Key
amount	Payment Amount
proof	Payment Proof
status	Payment Status

Documents Table

This table stores legal documents uploaded for each case.

Field	Description
document_id	Primary Key
case_id	Foreign Key
file_name	Document Name
file_path	File Storage Location

Audit Logs Table

This table records user activities performed within the system for monitoring and security purposes.

Field	Description
log_id	Primary Key
user_id	Foreign Key
activity	User Activity
timestamp	Date and Time of Activity

Notifications Table

This table stores notifications generated by the system for users.

Field	Description
notification_id	Primary Key
user_id	Foreign Key
message	Notification Content
status	Notification Status

Database Relationships

The relationships among these tables allow LEXSHIELD to manage user accounts, lawyer and client profiles, legal cases, appointments, messages, payments, documents, audit logs, and notifications. Foreign key relationships ensure that each record is properly associated with its corresponding user, client, lawyer, or case. This structure supports secure data

management, efficient retrieval of legal information, and reliable communication among system users while maintaining data integrity and consistency. The database also supports the implementation of security mechanisms such as role-based access control, secure authentication, activity logging, data encryption, and email verification to help protect sensitive legal information.

CHAPTER 5

CONCLUSION AND RECOMMENDATIONS

5.1 Introduction

This chapter presents the conclusions of the study entitled **LEXSHIELD: Integrated Cybersecurity and Legal Compliance Platform for Law Firms**. It summarizes the development of the system and explains how the objectives of the study were achieved. It also provides recommendations that may help improve the system and guide future researchers who plan to develop similar systems.

5.2 Summary of Findings

The researchers successfully developed LEXSHIELD, a web-based legal management system with built-in cybersecurity and legal compliance features for law firms. Based on the development of the system, the following findings were identified:

- The researchers developed a centralized platform that allows administrators, lawyers, clients, and public visitors to perform their assigned tasks through one system.
- The system provides important legal management features such as user registration, secure login, lawyer and client management, case management, appointment scheduling, document management, secure messaging, payment management, notifications, and audit logging.
- Security features such as user authentication, email verification, password encryption, role-based access control, CSRF protection, session guard, rate limiter, input validation,

data encryption, hashing, malware and phishing scanning, rate limiter and audit logging were included to help protect confidential legal information.

- The system was developed using the Agile Software Development Methodology, which allowed the researchers to build and improve each module step by step.
- A Three-Tier Client-Server Architecture and a MySQL database were used to organize data properly and support secure and efficient system operations.
- Different development tools such as PHP, HTML, CSS, JavaScript, MySQL, XAMPP, Composer, PHPMailer, Git, GitHub, and Visual Studio Code were successfully used to develop the system.

5.3 Conclusion

The study successfully developed **LEXSHIELD: Integrated Cybersecurity and Legal Compliance Platform for Law Firms**. The system combines legal management functions with cybersecurity features to help law firms manage legal information more securely and efficiently.

LEXSHIELD includes important features such as secure login, email verification, role-based access control, case management, appointment scheduling, document management, secure messaging, payment management, and audit logging. These features help protect sensitive legal information while making daily tasks easier to manage.

The use of the Agile Software Development Methodology helped the researchers develop the system in an organized way, while the Three-Tier Client-Server Architecture provided a secure and reliable structure for the application. Overall, the study achieved its goal of developing a legal management system that supports cybersecurity and legal compliance for law firms.

5.4 Recommendations

To further improve LEXSHIELD, the following recommendations are suggested:

System Improvements

- Add Two-Factor Authentication (2FA) to provide stronger account security.
- Add automatic security alerts to notify administrators of suspicious activities.
- Allow reports to be downloaded in PDF, Excel, and CSV formats.
- Improve the backup and recovery process to help prevent data loss.
- Add SMS or mobile notifications for appointments and important system updates.

Future Research

- Develop a mobile version of the system for easier access.
- Add Artificial Intelligence (AI) features that can help detect possible cybersecurity threats.
- Conduct further studies using more respondents to evaluate the system's performance and usability.
- Explore the integration of additional legal services, such as online consultations and digital signatures.

Implementation

- Law firms should provide training to users before using the system.
- Regular system maintenance, software updates, and database backups should be performed to keep the system secure and reliable.
- Administrators should regularly monitor user activities and audit logs to improve system security.

5.5 Impact of the Study

This study shows that legal management and cybersecurity can be combined into one web-based system. LEXSHIELD helps improve the management of legal information by providing secure access, organized records, and better protection of confidential data.

The study also provides a foundation for future researchers who want to develop similar systems for law firms or other organizations that manage sensitive information. Although the system is a prototype, it shows that technology can help improve legal services while supporting cybersecurity and data protection.

5.6 Summary

In summary, the researchers successfully developed **LEXSHIELD: Integrated Cybersecurity and Legal Compliance Platform for Law Firms**. The system provides useful legal management features together with cybersecurity measures that help protect sensitive information and improve daily operations.

The study achieved its goal of developing a secure and organized web-based platform for law firms. Future improvements and additional features can further increase the system's usefulness and support the changing needs of the legal profession.

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